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RynnBrain 1.1: Towards More Capable and Generalizable Embodied Foundation Model[a]

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We present RynnBrain 1.1, a family of embodied foundation models spanning 2B, 9B, and 122B-A10B scales. Trained with a unified spatio-temporal and physically grounded framework, RynnBrain 1.1 supports embodied perception, spatial reasoning, localization, and planning. Compared with RynnBrain 1.0, it further introduces contact-point prediction across the model family and native 3D grounding for the 2B and 9B models, yielding representations and outputs that are more directly aligned with robot manipulation. We also develop RynnBrain-VLA with a unified cross-embodiment action space and embodiment-specific masking, and deploy it on Unitree G1, Astribot-S1, and Tianji-Wuji. RynnBrain 1.1 achieves strong results on embodied cognition, localization, and 3D grounding, with the 122B-A10B model outperforming all evaluated proprietary and open-source models on VSI-Bench, MMSI, and RefSpatial-Bench. Real-robot experiments show that RynnBrain-initialized policies outperform Qwen-based and representative generalist VLAs, while joint multi-task and multi-embodiment training improves process scores and success rates over per-task training.

https://alibaba-damo-academy.github.io/RynnBrain
https://github.com/alibaba-damo-academy/RynnBrain
https://huggingface.co/collections/Alibaba-DAMO-Academy/rynnbrain-11

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[a]We would like to thank WUJI TECH for their generous and unconditional hardware support.

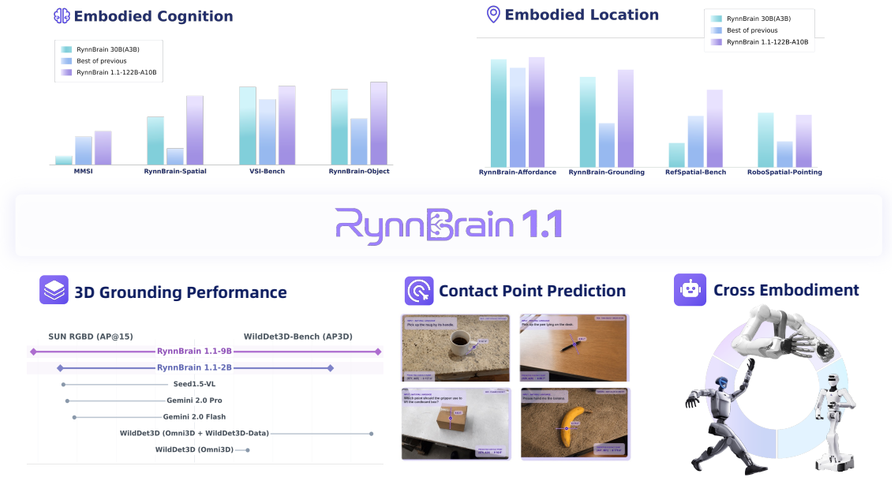

Figure 1 We introduce RynnBrain 1.1, a model family comprising three variants—2B, 9B, and 122B-A10B—together with the post-trained RynnBrain-VLA. RynnBrain 1.1 achieves leading performance across a broad suite of benchmarks. RynnBrain-VLA demonstrates significant promise in enabling action prediction across diverse embodiments within a unified action space.

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RynnBrain 1.1：迈向能力更强、泛化性更好的具身基础模型[a]

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我们提出 RynnBrain 1.1，这是一系列具身基础模型，规模覆盖 2B、9B 和 122B-A10B。RynnBrain 1.1 采用统一的时空与物理约束框架进行训练，支持具身感知、空间推理、定位与规划。与 RynnBrain 1.0 相比，它进一步在全系列模型中引入接触点预测，并为 2B 和 9B 模型引入原生 3D 定位能力，从而得到与机器人操作更直接对齐的表示和输出。我们还开发了 RynnBrain-VLA，具有统一的跨具身动作空间和具身特定掩码，并将其部署在 Unitree G1、Astribot-S1 和 Tianji-Wuji 上。RynnBrain 1.1 在具身认知、定位和 3D 定位方面取得了优异结果，其中 122B-A10B 模型在 VSI-Bench、MMSI 和 RefSpatial-Bench 上优于所有被评估的专有和开源模型。真实机器人实验表明，经 RynnBrain 初始化的策略优于基于 Qwen 的策略和有代表性的通用 VLA，而联合多任务与多具身训练相比逐任务训练提升了过程得分和成功率。

https://alibaba-damo-academy.github.io/RynnBrain
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日期：2026年7月31日

[a]我们衷心感谢 WUJI TECH 提供慷慨且无条件的硬件支持。

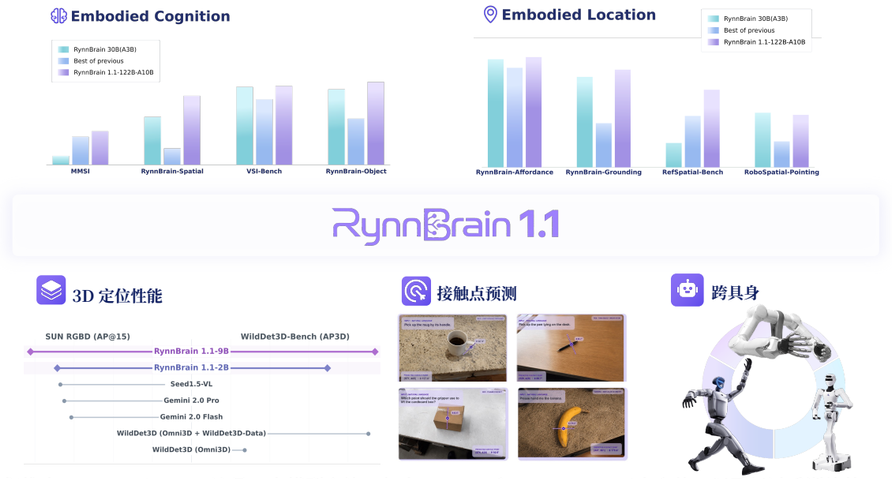

图1 我们推出 RynnBrain 1.1，这是一个模型家族，包含 2B、9B 和 122B-A10B 三个变体，以及经过后训练的 RynnBrain-VLA。RynnBrain 1.1 在广泛的基准测试中取得了领先性能。RynnBrain-VLA 在统一动作空间内实现跨多种具身形态的动作预测方面展现出巨大潜力。

1 Introduction

Embodied intelligence requires models to go beyond visual recognition and language understanding. A robot operating in the physical world must perceive objects [57, 55], reason about spatial relations [60, 80], identify actionable regions [27], and translate such understanding into executable behaviors [6, 5]. Although recent multimodal large language models have achieved strong performance in image and video question answering tasks [4, 76, 29, 17], real-world robotic systems impose additional requirements: they must understand 3D structure, reason across viewpoints, ground language instructions into physical locations, and support downstream manipulation policies. Therefore, the central challenge for embodied foundation models is not only whether they can understand visual content, but whether visual understanding is grounded in the physical world and, crucially, whether it can be transferred to real-robot action [64].

Recent embodied foundation models [28, 62, 20, 49, 65] have advanced rapidly along different axes, including object and region localization, spatio-temporal understanding, and robot-oriented planning. Specifically, RynnBrain 1.0 [20] explored this direction by introducing a unified embodied multimodal model for egocentric understanding, spatial grounding, physically grounded reasoning, and planning [33, 19, 52], demonstrating the feasibility of combining general and robot-oriented multimodal understanding. Despite its strong performance, several key aspects were not fully explored in RynnBrain 1.0. First, how can the representations and outputs from RynnBrain be brought closer to robot manipulation? Second, how effectively can RynnBrain serve as an initialization for downstream VLA post-training?

Motivated by these two questions, we develop RynnBrain 1.1, a new family of embodied foundation models upgraded from RynnBrain 1.0 [20]. Concretely, to learn representations that are more meaningful for robot manipulation, we add two new training tasks to the embodied pretraining of RynnBrain 1.1: (1) **contact point prediction** and (2) **3D grounding**. The contact point prediction task requires the model to locate the center point and the rotation angle of the gripper for grasping. Compared to the grasp rectangle detection task in [66, 20], contact point prediction can guide the model to provide more precise (i.e., the contact point) and meaningful (i.e., the planar rotation angle of the gripper) features for the downstream robot policy. As for 3D grounding, it introduces explicit and massive 3D modeling during the training of RynnBrain 1.1. Since robots operate in three-dimensional environments [67, 31], equipping embodied foundation models with explicit 3D modeling benefits manipulation through both richer representations and directly usable 3D outputs. In addition, to support the fine-tuning of RynnBrain 1.1 across heterogeneous robot embodiments and control interfaces, we introduce a unified action space that organizes embodiment-specific actions into semantically aligned body-part groups. Embodiment-specific masks are then applied to activate only the dimensions available on each robot. This design allows data from different embodiments to be trained jointly within a single policy without forcing explicit alignment between incompatible action spaces.

Built upon Qwen3.5 [53], RynnBrain 1.1 is available in three model scales—2B, 9B, and 122B-A10B—all trained under a unified capability definition and training framework. This consistent design enables us to systematically study how model capabilities evolve with scale. We evaluate RynnBrain 1.1 from two complementary perspectives. First, we benchmark the base models on a broad suite of embodied multimodal tasks covering spatio-temporal understanding, spatial reasoning, localization, and 3D grounding. The results show an upward trend from 2B to 9B and 122B-A10B, while the 2B and 9B models demonstrate strong language-conditioned 3D grounding after incorporating explicit 3D supervision. Second, we evaluate how large-scale embodied pretraining helps to learn downstream VLA policy. We perform evaluations of RynnBrain-VLA on three different platforms, a Unitree G1 humanoid[1], an Astribot-S1 bimanual robot[2], and a Tianji-Wuji dexterous-hand system[3]. Controlled evaluations on Astribot and Tianji-Wuji show that VLA policies built on our pretrained RynnBrain 1.1 outperform both the Qwen-based VLA policies and the most representative VLA generalists [6, 48]. Notably, joint multi-task and multi-embodiment training further improves the average process score and final success rate over separately fine-tuned per-task policies.

Overall, the contributions of this technical report are as follows:

- We release RynnBrain 1.1, a new family of embodied foundation models spanning 2B, 9B, and 122B-

[1] https://www.unitree.com/g1
[2] https://www.astribot.com/en/product
[3] https://www.tianjizn.com/products/marvin-series/, https://www.wuji.tech/zh/hand

1 引言

具身智能要求模型超越视觉识别和语言理解。在物理世界中运行的机器人必须感知物体 [57, 55], 推理空间关系 [60, 80], 识别可操作区域 [27], 并将这种理解转化为可执行行为 [6, 5]。尽管近期的多模态大语言模型在图像和视频问答任务中取得了出色表现 [4, 76, 29, 17], 真实世界的机器人系统仍提出了额外要求：它们必须理解三维结构、跨视角推理、将语言指令落地到物理位置，并支持下游操作策略。因此，具身基础模型的核心挑战不仅在于它们能否理解视觉内容，还在于视觉理解是否扎根于物理世界，并且更关键的是，它能否迁移到真实机器人动作 [64]。

近期，具身基础模型 [28, 62, 20, 49, 65] 已沿不同方向快速发展，包括目标和区域定位、时空理解以及面向机器人的规划。具体而言，RynnBrain 1.0 [20] 通过引入一个统一的具身多模态模型来探索这一方向，该模型用于第一人称理解、空间定位、基于物理的推理和规划 [33, 19, 52], ，展示了将通用多模态理解与面向机器人的多模态理解相结合的可行性。尽管 RynnBrain 1.0 表现出色，但若干关键方面尚未得到充分探索。首先，如何让 RynnBrain 的表示和输出更贴近机器人操作? 其次，RynnBrain 能在多大程度上有效地作为下游 VLA 后训练的初始化?

受这两个问题启发，我们开发了 RynnBrain 1.1，这是一个由 RynnBrain 1.0 [20] 升级而来的新具身基础模型系列。具体而言，为了学习对机器人操作更有意义的表示，我们在 RynnBrain 1.1 的具身预训练中新增了两个训练任务：（1）接触点预测和（2）3D 定位。接触点预测任务要求模型定位抓取时夹爪的中心点和旋转角度。与 [66, 20], 中的抓取矩形检测任务相比，接触点预测可以引导模型为下游机器人策略提供更精确（即接触点）且更有意义（即夹爪的平面旋转角度）的特征。至于 3D 定位，它在 RynnBrain 1.1 的训练过程中引入了显式且大规模的 3D 建模。由于机器人在三维环境中运行 [67, 31], ，为具身基础模型配备显式 3D 建模可通过更丰富的表示和可直接使用的 3D 输出，助力机器人操作。此外，为支持 RynnBrain 1.1 在异构机器人具身和控制接口上的微调，我们引入统一动作空间，将各具身特有的动作组织成语义对齐的身体部位组。随后应用各具身特定的掩码，仅激活每台机器人上可用的维度。该设计允许来自不同具身的数据在单一策略中联合训练，而无需在不兼容的动作空间之间强制显式对齐。

基于 Qwen3.5 [53]，构建的 RynnBrain 1.1 提供三种模型规模——2B、9B 和 122B-A10B——全部在统一的能力定义和训练框架下训练。这种一致的设计使我们能够系统地研究模型能力如何随规模演变。我们从两个互补的角度评估 RynnBrain 1.1。首先，我们在一套广泛的具身多模态任务上对基础模型进行基准测试，这些任务涵盖时空理解、空间推理、定位和三维空间定位。结果表明，从 2B 到 9B 再到 122B-A10B 呈上升趋势；而在引入显式 3D 监督后，2B 和 9B 模型展现出强大的语言条件下的三维空间定位能力。其次，我们评估大规模具身预训练如何帮助学习下游 VLA 策略。我们在三种不同平台上评估 RynnBrain-VLA：一台 Unitree G1 人形机器人[1]，一台 Astribot-S1 双臂机器人[2]，以及一套 Tianji-Wuji 灵巧手系统[3]。在 Astribot 和 Tianji-Wuji 上进行的受控评估表明，基于我们预训练的 RynnBrain 1.1 构建的 VLA 策略优于基于 Qwen 的 VLA 策略以及最具代表性的 VLA 通用模型 [6, 48]。值得注意的是，联合多任务与多具身训练相比分别微调的各任务策略，进一步提高了平均过程得分和最终成功率。

总体而言，本技术报告的贡献如下：

- 我们发布 RynnBrain 1.1，这是一个全新的具身基础模型系列，覆盖 2B、9B 和 122B-

[1] https://www.unitree.com/g1 [2] https://www.astribot.com/en/product [3] https://www.tianjizn.com/products/marvin-series/, https://www.wuji.tech/zh/hand

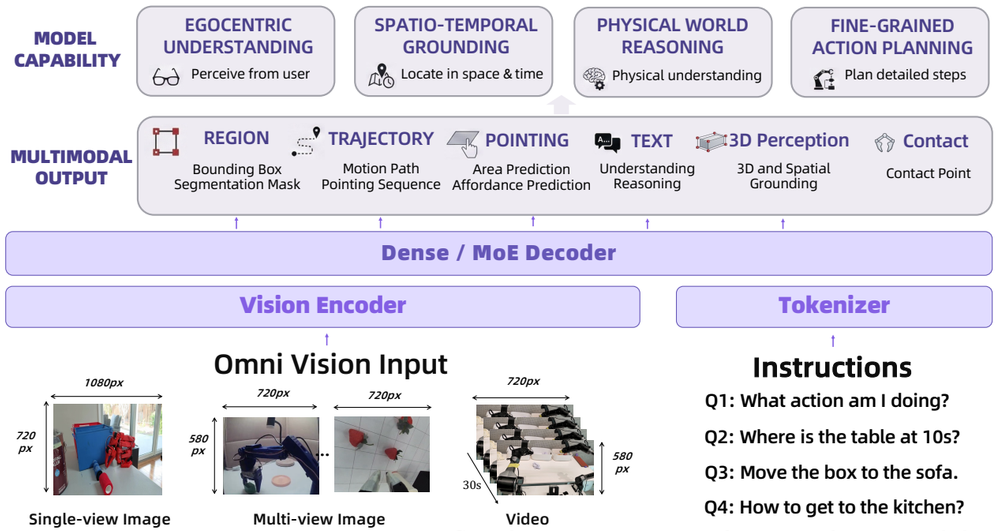

Figure 2 Overview of the RynnBrain 1.1 architecture. Given language instructions and omni-visual inputs—including single-view images, multi-view images, and videos—RynnBrain produces a unified set of multimodal outputs through either a dense or a mixture-of-experts decoder. These outputs cover text, regions, trajectories, pointing, 3D perception, and contact signals, enabling egocentric understanding, spatiotemporal grounding, physically grounded reasoning, and fine-grained action planning in real-world environments.

A10B parameter scales. Extensive evaluations across various benchmarks demonstrate a clear upward trend in model capabilities as the parameter scale increases. Notably, our RynnBrain 1.1-122B-A10B surpasses all proprietary models as well as open-source models on VSI-Bench, MMSI, and RefSpatial-Bench.

- We introduce two new features: (1) contact point prediction and (2) native 3D grounding (available for 2B and 9B-sized models) into embodied pretraining, yielding representations and outputs that are more directly aligned with robot manipulation.
- We develop RynnBrain-VLA, which couples a unified cross-embodiment action space with embodiment-specific masking. Beyond delivering strong real-world performance across heterogeneous embodiments, tasks, and control interfaces, it obtains further gains from joint multi-task and multi-embodiment training over separately fine-tuned per-task policies.

2 Overview

2.1 Model Architecture

An overview of the RynnBrain architecture is shown in Figure 2.

RynnBrain 1.1 adopts a decoder-only vision–language architecture following the design principles of Qwen3.5 [53]. It consists of a vision encoder, a vision–language projector, and a large language model (LLM) backbone initialized from Qwen3.5 variants. In this release, we build three model scales, namely RynnBrain 1.1-2B, RynnBrain 1.1-9B, and RynnBrain 1.1-122B-A10B. All models share the same overall architecture and training formulation, enabling a systematic study of embodied scaling across model sizes. We also adopt DeepStack [45] and Interleaved MRoPE [30] to better integrate multimodal information and support long-context spatio-temporal modeling over images, videos, and embodied observations.

图2 RynnBrain 1.1 架构概览。给定语言指令和全视觉输入——包括单视图图像、多视图图像和视频——RynnBrain 通过稠密解码器或混合专家解码器生成一组统一的多模态输出。这些输出涵盖文本、区域、轨迹、指向、3D 感知和接触信号，从而支持真实世界环境中的第一人称视角理解、时空定位、具有物理依据的推理以及细粒度动作规划。

A10B 参数规模。在各类基准上的广泛评估表明，随着参数规模增大，模型能力呈现出明显的上升趋势。值得注意的是，我们的 RynnBrain 1.1-122B-A10B 在 VSI-Bench、MMSI 和 RefSpatial- Bench 上超越了所有专有模型以及开源模型。

- 我们将两项新功能引入具身预训练：(1) 接触点预测；(2) 原生 3D 定位（适用于 2B 和 9B 规模的模型），从而产生与机器人操作更直接对齐的表征和输出。
- 我们开发了 RynnBrain-VLA，它将统一的跨具身动作空间与具身特定掩码相结合。除了在异构具身、任务和控制接口上展现出强大的真实世界性能外，它还能通过联合多任务和多具身训练，相较分别微调的逐任务策略获得进一步提升。

2 概述

2.1 模型架构

RynnBrain 架构概览如图 2 所示。

RynnBrain 1.1 采用仅解码器视觉-语言架构，遵循 Qwen3.5 [53] 的设计原则。它由视觉编码器、视觉-语言投影器和从 Qwen3.5 变体初始化的大型语言模型（LLM）主干组成。本次发布中，我们构建了三种模型规模，即 RynnBrain 1.1-2B、RynnBrain 1.1-9B 和 RynnBrain 1.1-122B-A10B。所有模型共享相同的整体架构和训练方案，从而能够系统研究跨模型规模的具身扩展。我们还采用 Deep-Stack [45] 和 Interleaved MRoPE [30]，以更好地整合多模态信息，并支持面向图像、视频和具身观测的长上下文时空建模。

2.2 Training Infrastructure

As described in RynnBrain [20], our training pipeline is designed for large-scale embodied multimodal data with highly heterogeneous sequence lengths. We follow the same infrastructure design, including online sequence-length-aware load balancing, per-sample loss reduction, memory-efficient training with ZeRO and gradient checkpointing, and optimized MoE execution with expert parallelism and efficient token dispatching. These components allow us to train on mixed image, video, text, and spatial-grounding data while mitigating straggler effects caused by long-tail sequence-length distributions.

For RynnBrain 1.1, we extend this infrastructure to the new Qwen3.5-based model family, covering 2B, 9B, and 122B scales. The same training pipeline is used across model sizes to ensure that differences in downstream performance mainly reflect model scaling and data updates rather than changes in infrastructure. For the 122B-A10B model, we adopt sparse MoE training with expert parallelism and communication-efficient token dispatching, following the infrastructure principles established in RynnBrain [20].

3 Pretraining

Similar to RynnBrain [20], our embodied pretraining framework is built around two key principles: (1) spatio-temporal memory and (2) physical-world grounding. Spatio-temporal memory enables the model to aggregate information from past visual observations, including objects, locations, events, and motion dynamics. Physical-world grounding, in turn, encourages the model to represent part of its understanding through explicit spatial outputs—such as objects, regions, affordances, and trajectories—rather than relying solely on unconstrained language generation. RynnBrain 1.1 advances this framework in two ways. First, we apply the same spatio-temporal embodied pretraining paradigm to three model scales—2B, 9B, and 122B-A10B—allowing us to examine how embodied perception, reasoning, localization, and planning change with model capacity. Second, as an initial step toward native 3D grounding, we add large-scale 3D-grounded data to the pretraining mixtures of RynnBrain 1.1-2B and RynnBrain 1.1-9B, enabling us to study the contribution of explicit 3D supervision to physical-space understanding at different model scales.

3.1 Training Recipe

Our pretraining formulation unifies temporal visual modeling and explicit spatial prediction within a single autoregressive framework. Multimodal observations are first encoded into a common token sequence, while both linguistic responses and grounded spatial targets are generated through the same prediction interface. This design equips RynnBrain with temporal memory and physically grounded representations without introducing separate task-specific decoders.

Unified Spatio-temporal Representation. A visual observation is represented as an ordered sequence of frames, $\mathbf{V} = \{I_t\}_{t=1}^{T}$. Static images correspond to the special case $T = 1$, whereas videos contain multiple temporally ordered frames with $T > 1$. For video inputs, we uniformly sample frames over time and encode each frame into visual tokens. Temporal position information is then attached to the resulting tokens so that the model can distinguish frame order and associate observations across time. By casting images and videos into the same sequential form, RynnBrain 1.1 can model object persistence, motion evolution, and long-range visual dependencies using a shared architecture.

Physically Grounded Output Space. Physical grounding is incorporated directly into the model's generation vocabulary. In addition to natural-language tokens, the output sequence may contain discrete representations of bounding boxes $\mathcal{B}$, points $\mathcal{P}$, and trajectory waypoints $\mathcal{T}$. Continuous image coordinates are first mapped to integers in $[0, 1000]$ and are then predicted autoregressively in the same manner as text tokens. As a result, semantic reasoning and spatial localization are learned jointly rather than handled by separate prediction heads. The 2B, 9B, and 122B-A10B models use this shared output specification in the main scaling experiments, while differences among them arise from model capacity and their corresponding training-data mixtures.

In addition, we introduce explicit 3D modeling task into the pretraining of RynnBrain 1.1-2B and RynnBrain 1.1-9B to explore native 3D grounding. These models follow the same instruction-following and autoregressive

2.2 训练基础设施

如 RynnBrain [20]，所述，我们的训练流水线面向序列长度高度异构的大规模具身多模态数据设计。我们沿用相同的基础设施设计，包括在线序列长度感知负载均衡、逐样本损失归约、使用 ZeRO 和梯度检查点的内存高效训练，以及借助专家并行和高效 token 调度优化的 MoE 执行。这些组件使我们能够在图像、视频、文本和空间定位混合数据上进行训练，同时缓解由长尾序列长度分布造成的掉队效应。

对于 RynnBrain 1.1，我们将这一基础设施扩展到基于 Qwen3.5 的新模型系列，覆盖 2B、9B 和 122B 规模。不同模型规模使用相同的训练流水线，以确保下游性能的差异主要反映模型扩展和数据更新，而非基础设施变化。对于 122B-A10B 模型，我们采用带专家并行和通信高效 token 分发的稀疏 MoE 训练，遵循 RynnBrain [20] 中确立的基础设施原则。

3 预训练

与 RynnBrain [20]，类似，我们的具身预训练框架围绕两项关键原则构建：（1）时空记忆；（2）物理世界锚定。时空记忆使模型能够聚合来自过去视觉观察的信息，包括物体、位置、事件和运动动态。物理世界锚定则鼓励模型通过显式空间输出——例如物体、区域、可供性和轨迹——来表达其部分理解，而不是仅依赖不受约束的语言生成。RynnBrain 1.1 从两方面推进了这一框架。首先，我们将相同的时空具身预训练范式应用于三种模型规模——2B、9B 和 122B-A10B——从而考察具身感知、推理、定位和规划如何随模型容量变化。其次，作为迈向原生 3D 锚定的初步步骤，我们在 RynnBrain 1.1-2B 和 RynnBrain 1.1-9B 的预训练数据混合中加入了大规模 3D 锚定数据，使我们能够研究显式 3D 监督在不同模型规模下对物理空间理解的贡献。

3.1 训练配方

我们的预训练方案在单一自回归框架内统一了时序视觉建模与显式空间预测。多模态观测首先被编码为统一的 token 序列，而语言响应和具有物理依据的空间目标均通过相同的预测接口生成。这一设计使 RynnBrain 具备时序记忆和具有物理依据的表征，而无需引入独立的任务专用解码器。

统一的时空表征。视觉观测被表示为有序帧序列 $\mathrm{V} = \{I_t\}_{t=1}^{T}$。静态图像对应特殊情况 $T = 1$，而视频则包含多个按时间排序且满足 $T > 1$ 的帧。对于视频输入，我们沿时间维度均匀采样帧，并将每一帧编码为视觉 token。随后将时间位置信息附加到所得 token 上，使模型能够区分帧顺序并关联跨时间的观测。通过将图像和视频转换为相同的序列形式，RynnBrain 1.1 能够使用共享架构对对象持续性、运动演化以及长程视觉依赖进行建模。

物理锚定的输出空间。物理锚定被直接纳入模型的生成词表。除了自然语言 token 外，输出序列还可以包含边界框 $\mathcal{B}$、点 $\mathcal{P}$ 和轨迹路点 $\mathcal{T}$ 的离散表示。连续图像坐标首先被映射为 $[0, 1000]$ 中的整数，然后以与文本 token 相同的方式自回归预测。因此，语义推理与空间定位是联合学习的，而不是由独立的预测头分别处理。2B、9B 和 122B-A10B 模型在主要规模扩展实验中使用这一共享输出规范，而它们之间的差异源于模型容量及其对应的训练数据混合配比。

此外，我们在 RynnBrain 1.1-2B 和 RynnBrain 1.1-9B 的预训练中引入显式 3D 建模任务，以探索原生 3D 定位。这些模型遵循相同的指令跟随和自回归

training paradigm as the rest of the RynnBrain 1.1 family, while extending the supervision from image-plane grounding toward physical-space understanding.

Specifically, the model takes as input a natural language instruction and camera intrinsics, and is trained to directly predict a 3D bounding box parameterized in the camera coordinate system—including center position $(\mathrm{cx}, \mathrm{cy}, \mathrm{cz})$, box dimensions $(\mathrm{w}, \mathrm{l}, \mathrm{h})$, and orientation $(\mathrm{pitch}, \mathrm{yaw}, \mathrm{roll})$, all in physical units (meters and radians). This 9-dimensional output format follows the design of prior works such as Seed-VL [58] and Qwen3-VL [4], which also adopt a compact set of numerical parameters for spatially grounded predictions. To preserve the unified autoregressive framework, we discretize all continuous 3D quantities into integer tokens within a fixed range, analogous to our 2D spatial tokenization. This design enables the model to perform physically grounded reasoning from a single image, inferring not only what and where in the pixel plane, but also how objects are positioned and oriented in real-world 3D space, directly bridging visual perception and downstream robotic or navigation tasks.

Optimization. All RynnBrain 1.1 variants are optimized end-to-end with a causal autoregressive objective over sequences containing both language and spatial tokens. Given a visual input $\mathbf{V}$ and a target sequence $\mathbf{y} = \{y_i\}_{i=1}^{L}$, the model minimizes the negative log-likelihood

$$\mathcal{L} = -\sum_{i=1}^{L} \log P\left(y_i \mid y_{<i}, \mathbf{V}; \mathbf{\Theta}\right). \tag{1}$$

Here, L denotes the target-sequence length and $\mathbf{\Theta}$ denotes the learnable model parameters. The same optimization objective is used for the entire RynnBrain 1.1 family, including the 3D-grounded 2B and 9B variants. We select scale-specific hyperparameters through preliminary experiments on representative subsets of the pretraining corpus, with the complete configurations summarized in Table 1.

Table 1 Hyperparameters of the pretraining stage for the RynnBrain model series.

Parameter	RynnBrain 1.1-2B	RynnBrain 1.1-9B	RynnBrain 1.1-122B-A10B
Base Model	Qwen3.5-2B	Qwen3.5-9B	Qwen3.5-122B-A10B
Optimizer	AdamW	AdamW	AdamW
Learning Rate	5×10^{-6}	2×10^{-6}	2×10^{-6}
Learning Rate Vision	1×10^{-6}	2×10^{-6}	2×10^{-6}
Global Batch Size	512	1024	1024
Warmup Ratio	0.03	0.03	0.03

3.2 Pretraining Data

RynnBrain 1.1 follows the capability-oriented data organization of RynnBrain [20]. As shown in Table 2, we organize the pretraining mixture into general multimodal data, cognition data, spatio-temporal localization data, 3D-grounded data, robot-oriented perception data, and planning data. For data types already introduced in RynnBrain 1.0, we follow the same instruction-following format and data processing protocol. Specifically, visual inputs are represented as images or videos, and model outputs are converted into mixed sequences of text and spatial tokens, including boxes, points, trajectories, and grasp-related annotations. In this section, we focus on the data updates introduced in RynnBrain 1.1.

General MLLM Data. To preserve broad multimodal understanding, we continue to use a general-purpose image–video–text corpus covering image understanding, video understanding, visual question answering, temporal reasoning, and instruction following. The mixture includes commonly used MLLM datasets such as LLaVA-OV-SI [37], LLaVA-Video [78], ShareGPT-4o-video [13], VideoGPT-plus [43], FineVideo [24], CinePile [56], ActivityNet [8], YouCook2 [81], and LLaVA-SFT [40]. In RynnBrain 1.1, we further incorporate self-collected RynnBrain-Thinking data. RynnBrain-Thinking is included as part of the general mixture to better align the model with the Qwen3.5-Thinking style. All data are converted into the unified instruction-following format before being mixed for pretraining.

训练范式与 RynnBrain 1.1 家族的其他模型一致，同时将监督从图像平面定位扩展到物理空间理解。

具体而言，该模型将自然语言指令和相机内参作为输入，并经过训练直接预测在相机坐标系中参数化的 3 D 边界框——包括中心位置 (cx, cy, cz)、边界框尺寸 (w, l, h) 和朝向 (pitch, yaw, roll)，全部采用物理单位（米和弧度）。这种 9 维输出格式沿用了 Seed-VL [58] 和 Qwen3-VL [4], 等先前工作的设计，这些工作同样采用一组紧凑的数值参数进行空间定位预测。为保持统一的自回归框架，我们将所有连续 3D 量在固定范围内离散为整数 token，类似于我们的 2D 空间 token 化。这一设计使模型能够基于单张图像进行具有物理依据的推理，不仅推断像素平面中的“是什么”和“在哪里”，还能推断物体在真实世界 3D 空间中的位置与朝向，从而直接衔接视觉感知与下游机器人或导航任务。

优化。所有 RynnBrain 1.1 变体均通过因果自回归目标，在同时包含语言 token 与空间 token 的序列上进行端到端优化。给定视觉输入 V 和目标序列 y $= \{y_i\}_{i=1}^{L}$，模型最小化负对数似然

$$\mathcal{L} = -\sum_{i=1}^{L} \log P\left(y_i \mid y_{<i}, \mathbf{V}; \mathbf{\Theta}\right). \tag{1}$$

其中， L表示目标序列长度， $\mathbf{\Theta}$表示可学习的模型参数。整个 RynnBrain 1.1 系列均使用相同的优化目标，包括具备 3D 定位能力的 2B 和 9B 变体。我们通过在预训练语料的代表性子集上进行初步实验，选择各规模专属的超参数，完整配置汇总于表 1。

表1 RynnBrain 模型系列预训练阶段的超参数。

参数	RynnBrain 1.1-2B	RynnBrain 1.1-9B	RynnBrain 1.1-122B-A10B
基础模型	Qwen3.5-2B	Qwen3.5-9B	Qwen3.5-122B-A10B
优化器	AdamW	AdamW	AdamW
学习率	5×10^{-6}	2×10^{-6}	2×10^{-6}
视觉学习率	1×10^{-6}	2×10^{-6}	2×10^{-6}
全局批次大小	512	1024	1024
预热比例	0.03	0.03	0.03

3.2 预训练数据

RynnBrain 1.1 沿用了 RynnBrain [20] 以能力为导向的数据组织方式。如表 2 所示，我们将预训练数据混合组织为通用多模态数据、认知数据、时空定位数据、三维场景关联数据、面向机器人的感知数据以及规划数据。对于 RynnBrain 1.0 中已介绍的数据类型，我们沿用相同的指令遵循格式和数据处理协议。具体而言，视觉输入表示为图像或视频，模型输出则转换为文本与空间 token 的混合序列，包括框、点、轨迹以及抓取相关标注。本节重点介绍 RynnBrain 1.1 中引入的数据更新。

通用 MLLM 数据。为保持广泛的多模态理解能力，我们继续使用通用图像—视频—文本语料库，覆盖图像理解、视频理解、视觉问答、时序推理和指令遵循。该混合数据包括常用的 MLLM 数据集，例如 LLaVA-OV-SI [37], 、LLaVA-Video [78], 、ShareGPT-4o-video [13], 、VideoGPT-plus [43], 、FineVideo [24],、CinePile [56], 、ActivityNet [8], 、YouCook2 [81], 和 LLaVA-SFT [40]。在 RynnBrain 1.1 中，我们进一步纳入自收集的 RynnBrain-Thinking 数据。RynnBrain-Thinking 作为通用混合数据的一部分被纳入，以更好地使模型对齐 Qwen3.5-Thinking 风格。所有数据在混合用于预训练之前，均被转换为统一的指令遵循格式。

Table 2 Pretraining data mixture for RynnBrain 1.1. New or substantially expanded data sources compared with RynnBrain 1.0 are highlighted in blue.

Category	Sub-Task	Data Sources
General MLLM	General	LLaVA-OV-SI [37], LLaVA-Video [78], ShareGPT-4o-video [13], VideoGPT-plus [43], FineVideo [24], CinePile [56], ActivityNet [8], YouCook2 [81], LLaVA-SFT [40], VL3 [76], RynnBrain-Thinking
Cognition	Object Understanding	RynnBrain-Object, RefCOCO [72], Google Refexp [44], Osprey-724K [74], DAM [39], VideoRefer-700k [75]
	Spatial Understanding	SenseNova-SI-800K [10], VSI-590K [69], VLM-3R [22], RynnBrain-Spatial
	Counting	RynnBrain-Counting, Molmo2 [15]
	OCR	RynnBrain-OCR, Llama-Nemotron-VLM OCR [47]
	Egocentric Task Understanding	EgoRe-5M [52], EgoTaskQA [33], Env-QA [25], QAEgo4D [26], RoboVQA [59], Robo2VLM [12], ShareRobot [32]
Localization	Object Localization	ADE20K [79], COCOStuff [9], Mapillary [46], PACO-LVIS [54], PASCAL-Part [14], VG [35], RoboAfford-Object [27], RynnBrain-Grounding
	Area Localization	RefSpatial [80], RoboAfford-Area [27], Molmo2 [15], RynnBrain-Area, RoboRefer [80],
	Affordance Localization	RynnBrain-Affordance, RoboAfford-Affordance [27]
	Trajectory Prediction	RynnBrain-Trajectory, FSD [73], MolmoAct [36]
	3D Grounding	WildDet3D Essential [31], WildDet3D Synthetic [31], FoundationPose [67]
	Grasp / Contact Point Prediction	Grasp-Anything-Visible [66], GraspClutter6D [3], GraspNet-1B [23], Jacquard V2 [38], GraspFactory [61]
Planning	Manipulation	AgibotWorld [18], Open X-Embodiment [16], RynnBrain-Planning, Galaxea-G0 [34]

Multi-dimensional Cognition Data. For embodied cognition, RynnBrain 1.1 follows the task taxonomy of RynnBrain [20], covering object understanding, spatial understanding, counting, OCR, and egocentric task understanding. Object understanding data include RynnBrain-Object, RefCOCO [72], Google Refexp [44], Osprey-724K [74], DAM [39], and VideoRefer-700K [75], which provide object-centric supervision for fine-grained recognition and reasoning. Spatial understanding data include SenseNova-SI-800K [10], VSI-590K [69], VLM-3R [22], and RynnBrain-Spatial, supporting reasoning over spatial relations, viewpoints, and scene structure. For counting and OCR, we use RynnBrain-Counting, Molmo2 [15], RynnBrain-OCR, and the newly added Llama-Nemotron-VLM OCR data. For egocentric task understanding, we use EgoRe-5M [52], EgoTaskQA [33], Env-QA [25], QAEgo4D [26], RoboVQA [59], Robo2VLM [12], and ShareRobot [32]. These datasets are converted following the RynnBrain instruction format to provide broad supervision for object-centric, spatial, textual, and first-person task understanding.

Spatio-temporal Localization Data. RynnBrain 1.1 inherits the localization formulation of RynnBrain 1.0, where spatial outputs are represented using normalized coordinate tokens. The localization mixture covers object localization, area localization, affordance localization, trajectory prediction, and grasp/contact-related prediction. Object localization uses ADE20K [79], COCOStuff [9], Mapillary [46], PACO-LVIS [54], PASCAL-Part [14], VG [35], RoboAfford-Object [27], and RynnBrain-Grounding. Area localization uses RefSpatial [80], RoboAfford-Area [27], Molmo2 [15], RynnBrain-Area, and the newly added RoboRefer. Affordance localization uses RynnBrain-Affordance and RoboAfford-Affordance [27]. Trajectory prediction uses RynnBrain-Trajectory, FSD [73], and the newly added MolmoAct. For these data sources, we follow the RynnBrain processing protocol and convert different annotations into a shared output space consisting of boxes, points, and trajectories.

表2 预训练数据混合用于 RynnBrain 1.1。新增或大幅扩展的数据来源，与 RynnBrain 1.0 相比，被高亮显示为蓝色。

类别	子任务	数据来源
通用多模态大模型	通用	LLaVA-OV-SI [37], LLaVA-Video [78], ShareGPT-4o-video [13], VideoGPT-plus [43],FineVideo [24], CinePile [56], ActivityNet [8], YouCook2 [81], LLaVA-SFT [40], VL3 [76],RynnBrain-Thinking
认知	物体理解	RynnBrain-Object, RefCOCO [72],Google Refexp [44], Osprey-724K [74], DAM [39],VideoRefer-700k [75]
	空间理解	SenseNova-SI-800K [10],VSI-590K [69], VLM-3R [22], RynnBrain-Spatial
	计数	RynnBrain-Counting、Molmo2 [15]
	OCR	RynnBrain-OCR、Llama-Nemotron-VLM OCR [47]
	第一人称视角任务理解	EgoRe-5M [52],EgoTaskQA [33], Env-QA [25], QAEgo4D [26], RoboVQA [59], Robo2VLM [12], ShareRobot [32]
定位	物体定位	ADE20K [79], COCOStuff [9],Mapillary [46], PACO-LVIS [54], PASCAL-Part [14], VG [35], RoboAfford-Object [27],RynnBrain-Grounding
	区域定位	RefSpatial [80], RoboAfford-Area [27], Molmo2 [15], RynnBrain-Area，RoboRefer [80],
	可供性定位	RynnBrain-Affordance, RoboAfford-Affordance [27]
	轨迹预测	RynnBrain-Trajectory, FSD [73], MolmoAct [36]
	3D 定位	WildDet3D 基础版 [31], WildDet3D 合成版 [31], FoundationPose [67]
	抓取 / 接触点预测	Grasp-Anything-Visible [66], GraspClutter6D [3], GraspNet-1B [23], Jacquard V2 [38],GraspFactory [61]
规划	操作	AgibotWorld [18],Open X-Embodiment [16], RynnBrain-规划，Galaxea-G0 [34]

多维认知数据。在具身认知方面，RynnBrain 1.1 沿用 RynnBrain [20]，的任务分类体系，涵盖物体理解、空间理解、计数、OCR 和自我中心任务理解。物体理解数据包括 RynnBrain-Object、RefCOCO [72]，、Google Refexp [44]，、Osprey-724K [74]，、DAM [39]，和 VideoRefer-700K [75]，，这些数据为细粒度识别与推理提供以物体为中心的监督。空间理解数据包括 SenseNova-SI-800K [10]，、VSI-590K [69]，、VLM-3R [22]，和 RynnBrain-Spatial，支持对空间关系、视角和场景结构进行推理。对于计数和 OCR，我们使用 RynnBrain-Counting、Molmo2 [15]，、RynnBrain-OCR 以及新增的 Llama-Nemotron-VLM OCR 数据。对于自我中心任务理解，我们使用 EgoRe-5M [52]，、EgoTaskQA [33]，、Env-QA [25]，、QAEgo4D [26]，、RoboVQA [59]，、Robo2VLM [12]，和 ShareRobot [32]。这些数据集均按照 RynnBrain 指令格式进行转换，从而为以物体为中心、空间、文本和第一人称任务理解提供广泛监督。

时空定位数据。RynnBrain 1.1 沿用了 RynnBrain 1.0 的定位表示方案，其中空间输出使用归一化坐标 token 表示。定位数据混合涵盖对象定位、区域定位、可供性定位、轨迹预测以及抓取/接触相关预测。对象定位使用 ADE20K [79], COCOStuff [9], Mapillary [46], PACO-LVIS [54],PASCAL-Part [14], VG [35], RoboAfford-Object [27], 和 RynnBrain-Grounding。区域定位使用 RefSpatial [80], RoboAfford-Area [27], Molmo2 [15], RynnBrain-Area，以及新增的 RoboRefer。可供性定位使用 RynnBrain-Affordance 和 RoboAfford-Affordance [27]。轨迹预测使用 RynnBrain-Trajectory、FSD [73], 以及新增的 MolmoAct。对于这些数据源，我们遵循 RynnBrain 处理协议，并将不同标注转换为由框、点和轨迹构成的共享输出空间。

3D-grounded Data. To explore native 3D grounding, we conduct a 3D-grounded preview training in the 2B and 9B setting by mixing explicit 3D annotations into the pretraining data. Given an image, a natural language instruction, and camera intrinsics, the model is trained to predict a 3D bounding box in the camera coordinate system. The box is parameterized by its center position $(\mathrm{cx}, \mathrm{cy}, \mathrm{cz})$, dimensions $(\mathrm{w}, \mathrm{l}, \mathrm{h})$, and orientation $(\mathrm{pitch}, \mathrm{yaw}, \mathrm{roll})$, represented in physical units such as meters and radians.

To support this at scale, we construct a multi-source corpus combining large-scale automatically lifted annotations with high-fidelity synthetic data. For broad category and scene coverage, we adopt **WildDet3D-Data** [31], a recently proposed open-vocabulary 3D detection dataset that lifts 2D annotations from COCO, LVIS, Objects365, and V3Det into 3D via a multi-model candidate generation and selection pipeline. We use its two curated subsets after additional filtering: *Essential* (102,979 images, 374K manually verified annotations after filtering out geometrically implausible candidates and small objects, 12,064 categories) and *Synthetic* (896,004 images, 888K automatically selected annotations, 11,896 categories, filtered by a fine-tuned Molmo2 scoring six perceptual criteria with a retention threshold >10), offering unprecedented category diversity but inheriting residual noise from monocular lifting. For each 3D annotation, we leverage Qwen-VL to generate an open-ended referring query based on the corresponding 2D bounding box and category label, incorporating spatial descriptions (e.g., relative positions such as "left," "right," "front," or "behind") and attribute modifiers (e.g., color, size, or material cues) to make the referring expression more discriminative and precise for grounding. To complement this with physically exact supervision, we additionally incorporate **FoundationPose** [67], a synthetic dataset derived from the Google Scanned Objects subset under controlled camera trajectories, which provides 1,849K images with noise-free 3D bounding boxes computed directly from ground-truth object-to-camera transformations, with each asset assigned a natural-language category via Qwen-VL to bridge to real-world semantics. Though its category coverage is limited to 928 instances, its metric-scale accuracy and zero-depth-ambiguity geometry serve as a strong regularizer for depth and orientation prediction, teaching correct inductive biases for camera-to-world projection when mixed with the diverse but noisier WildDet3D annotations.

Contact Point Data. RynnBrain 1.0 represented planar grasps using the four corners of an oriented rectangle. We found this detector-style representation poorly aligned with action grounding in a general embodied model. A target object typically admits a set of functionally valid grasps rather than a unique rectangle. Moreover, unlike an object bounding box, a grasp rectangle has no canonical image-space extent: its width, height, aspect ratio, and boundary placement depend on gripper geometry and dataset-specific annotation conventions. Consequently, IoU against a single reference rectangle may penalize functionally equivalent grasps and is not necessarily correlated with execution success. The four-corner representation also introduces redundant extent variables and geometric consistency constraints into autoregressive generation. RynnBrain 1.1 therefore adopts a compact contact-centered representation $(a = (p, \theta))$, where $(p = (x, y))$ denotes the center of the annotated grasp configuration and (θ) is the in-plane angle of the line connecting the gripper fingers. The coordinates are normalized to [0,1000], and each target is serialized as <grasp pose> (x, y), θ </grasp pose>. This representation factors out annotation- and embodiment-dependent rectangle extents while preserving an action-relevant spatial anchor and orientation cue.

A substantial portion of our source data consists of simulation-native, multidimensional grasp annotations rather than directly usable image–text pairs. We therefore develop a view-conditioned rendering and projection pipeline to convert these annotations into reliable 2D supervision. For each object–grasp configuration, we select a camera pose subject to multiple observability constraints: the target object must remain fully inside the image, occupy an appropriate range of image scales, and avoid severe truncation or occlusion; meanwhile, the projected grasp center and orientation must both remain visible and spatially distinguishable. Views in which the object is excessively near, distant, truncated, or visually ambiguous, or in which the grasp configuration is not observable, are discarded. The retained views are rendered as RGB images, after which the multidimensional grasp annotations are projected into the common $((p, \theta))$ representation. This procedure is essential because unconstrained camera sampling can produce geometrically valid annotations that cannot be inferred from the rendered image.

After rendering, projection, and filtering, the resulting corpus contains 2.6M training samples from Grasp-Anything [66], Jacquard V2 [38], GraspFactory [61], GraspNet-1B [23], and GraspClutter6D [3]. We treat $((p, \theta))$ as an action-grounding interface rather than a complete executable grasp pose; depth, approach

3D 接地数据。为探索原生 3D 接地能力，我们在 2B 和 9B 设置下通过将显式 3D 标注混入预训练数据来进行 3D 接地预览训练。给定一张图像、自然语言指令和相机内参，训练模型预测相机坐标系中的 3D 边界框。该框由其中心位置 (cx, cy, cz)、尺寸 (w, l, h) 和朝向 (pitch、yaw、roll) 参数化，并以米和弧度等物理单位表示。

为了在规模化层面支持这一点，我们构建了一个多源语料库，将大规模自动提升的标注与高保真合成数据相结合。为了获得广泛的类别和场景覆盖，我们采用 WildDet3D-Data [31],——一个近期提出的开放词汇 3D 检测数据集，它通过多模型候选生成与选择流程，将来自 COCO、LVIS、Objects365 和 V3Det 的 2D 标注提升到 3D。经过额外过滤后，我们使用其两个精选子集：Essential（102,979 张图像，在过滤掉几何上不合理的候选和小物体后，有 374K 条人工验证标注，12,064 个类别）和 Synthetic（896,004 张图像，888K 条自动选中的标注，11,896 个类别，由微调后的 Molmo2 按六个感知标准评分并以保留阈值 >10 过滤），提供了前所未有的类别多样性，但也继承了单目提升带来的残余噪声。对于每个 3D 标注，我们利用 Qwen-VL 根据对应的 2D 边界框和类别标签生成开放式指代查询，并纳入空间描述（例如“左”“右”“前”“后”等相对位置）和属性修饰语（例如颜色、大小或材质线索），使指代表达更具区分度、更精确，便于定位。为了用物理上精确的监督来补充这一点，我们还引入了 FoundationPose [67], ——一个源自 Google Scanned Objects 子集、在受控相机轨迹下生成的合成数据集，它提供 1,849K 张图像以及无噪声的 3D 边界框；这些边界框直接由真值物体到相机的变换计算得到，每个资产都通过 Qwen-VL 分配自然语言类别，以桥接到真实世界语义。尽管其类别覆盖范围仅限于 928 个实例，但其公制尺度精度和无深度歧义的几何结构可作为深度和朝向预测的强正则化项，在与多样但噪声更大的 WildDet3D 标注混合时，教会模型正确的相机到世界投影归纳偏置。

接触点数据。RynnBrain 1.0 使用定向矩形的四个角来表示平面抓取。我们发现，这种检测器式表示与通用具身模型中的动作接地对齐不佳。目标物体通常允许一组功能上有效的抓取，而不是一个唯一矩形。此外，与物体边界框不同，抓取矩形没有规范的图像空间范围：其宽度、高度、长宽比和边界位置取决于夹爪几何结构和数据集特定的标注约定。因此，与单一参考矩形计算 IoU 可能会惩罚功能等价的抓取，并且不一定与执行成功相关。四角表示还会将冗余的范围变量和几何一致性约束引入自回归生成。因此，RynnBrain 1.1 采用一种紧凑的以接触点为中心的表示 $(a = (p, \theta))$，其中 $(p = (x, y))$ 表示标注抓取配置的中心，(θ) 是夹爪手指连线在平面内的角度。坐标归一化到 [0,1000], ，并且每个目标序列化为 <抓取位姿> (x, y)，θ</抓取位姿>。该表示将依赖于标注和具身形态的矩形范围因素排除在外，同时保留与动作相关的空间锚点和方向线索。

我们的源数据中有相当一部分由仿真原生的多维抓取标注组成，而非可直接使用的图文对。因此，我们开发了一种基于视角的渲染与投影流程，将这些标注转换为可靠的二维监督信号。对于每个物体-抓取配置，我们在多个可观测性约束下选择相机位姿：目标物体必须完整位于图像内，占据合适的图像尺度范围，并避免严重截断或遮挡；同时，投影后的抓取中心和朝向必须均保持可见且在空间上可区分。对于物体过近、过远、被截断或视觉上存在歧义，或者抓取配置不可观测的视角，我们将其丢弃。保留的视角被渲染为 RGB 图像，随后多维抓取标注被投影到共同的 $((p, \theta))$ 表示中。这一过程至关重要，因为无约束的相机采样可能产生在几何上有效但无法从渲染图像中推断的标注。

经过渲染、投影和过滤后，所得语料库包含来自 Grasp- Anything [66], 、Jacquard V2 [38], 、GraspFactory [61], 、GraspNet-1B [23], 和 GraspClutter6D [3] 的 260 万个训练样本。我们将 $((p, \theta))$ 视为一种动作接地接口，而非完整的可执行抓取位姿；深度、接近

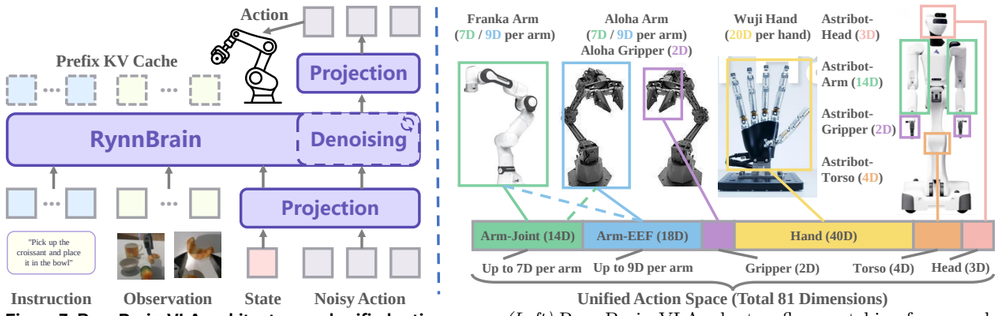

Figure 3 RynnBrain-VLA architecture and unified action space. *(Left)* RynnBrain-VLA adopts a flow-matching framework in which the RynnBrain base model serves as a single-stream Diffusion Transformer (DiT). The model takes a packed sequence of language instruction, multi-camera observations, robot state, and noisy action chunks as input, and iteratively denoises to predict an action chunk. The instruction token prefix is cached via KV cache to reduce inference latency. *(Right)* Actions from heterogeneous robot embodiments are represented within a shared 81-dimensional action space partitioned into semantically aligned body-part groups: Arm-Joint (14D, up to 7D per arm), Arm-EEF (18D, up to 9D per arm), Gripper (2D), Hand (40D), Torso (4D), and Head (3D). Each embodiment activates only its physically available dimensions via an embodiment-specific mask, enabling joint training across robots with incompatible low-level action spaces. In our evaluated embodiments: Unitree G1 activates Hand (14D, 7D per hand) within the unified space alongside a separately predicted 64D SONIC [42] latent token; Tianji-Wuji (Tianji Arm + Wuji Hand) activates Arm-Joint (14D) and Hand (40D); Astribot S1 activates Arm-Joint (14D), Gripper (2D), Head (3D) and Torso (4D). The robots shown in the figure are illustrative examples of the masking mechanism.

direction, gripper aperture, collision constraints, and robot kinematics are resolved by the downstream manipulation policy.

Physics-aware Planning Data. For manipulation planning, RynnBrain 1.1 follows the structured planning format of RynnBrain [20], where high-level task instructions are paired with grounded sub-task descriptions involving objects, areas, and affordances. The planning mixture includes AgibotWorld [18], Open X-Embodiment [16], RynnBrain-Planning, and the newly added Galaxea-G0. For existing grounded planning data, we follow the same annotation format and conversion protocol as RynnBrain 1.0. For Galaxea-G0, the original annotations provide sub-task-level labels but do not contain explicit high-level task instructions. We therefore construct the corresponding global task descriptions using Gemini 3.1 Pro, conditioned on the annotation IDs and the sequence of annotated sub-tasks. The resulting data are then converted into the same grounded planning format as the rest of the planning mixture.

4 Post-training for Vision-Language-Action Model

We use downstream VLA post-training as both a practical capability and a probe of base-model quality. This section describes the VLA *method*: the policy architecture (Section 4.1), the unified action space design (Section 4.2), the cross-embodiment / cross-driver deployment framework (Section 4.3), Real-Time Chunking for smooth low-latency control (Section 4.4), and the fine-tuning setup shared by all backbones (Section 4.5). The corresponding real-robot results are reported in Section 5.6.

4.1 Model Architecture

As shown in Figure 3, the model architecture of **RynnBrain-VLA** in RynnBrain-1.1 largely follows that in RynnBrain-1.0 [20], where the VLM backbone itself serves as a single-stream diffusion transformer and processes dense interaction among vision, language, and action in each layer. A flow matching framework is adopted to predict an action chunk [5] at each step. To preserve the VLM's inherent instruction-following capabilities, we utilize its native conversation format for organizing the input:

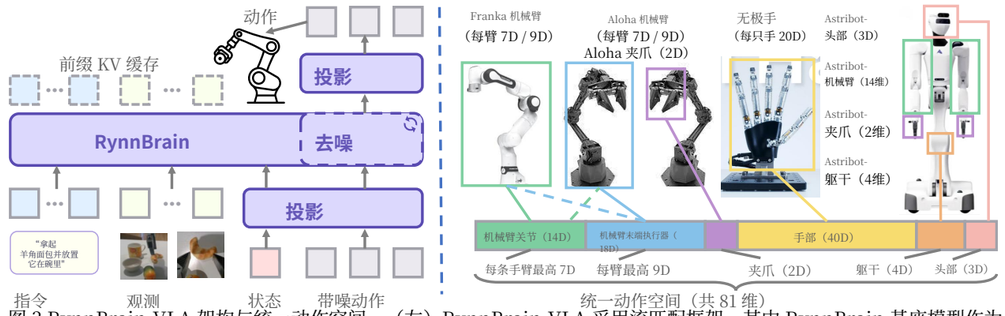

图 3 RynnBrain-VLA 架构与统一动作空间。（左）RynnBrain-VLA 采用流匹配框架，其中 RynnBrain 基座模型作为单流 Diffusion Transformer（DiT）。该模型以打包序列为输入，该序列包含语言指令、多相机观测、机器人状态和带噪动作块，并通过迭代去噪预测动作块。指令 token 前缀通过 KV 缓存进行缓存，以减少推理延迟。（右）来自异构机器人本体的动作在共享的 81 维动作空间中表示，该空间划分为语义对齐的身体部位组：Arm-Joint（14D，每臂最高 7D）、Arm-EEF（18D，每臂最高 9D）、Gripper（2D）、Hand（40D）、Torso（4D）和 Head（3D）。每个本体仅通过本体特定掩码激活其物理可用维度，从而支持在底层动作空间不兼容的机器人之间进行联合训练。在我们评估的本体中：Unitree G1 在统一空间内激活 Hand（14D，每只手 7D），并同时使用单独预测的 64D SONIC [42] 潜在 token；Tianji-Wuji（Tianji Arm + Wuji Hand）激活 Arm-Joint（14D）和 Hand（40D）；Astribot S1 激活 Arm-Joint（14D）、Gripper（2D）、Head（3D）和 Torso（4D）。图中展示的机器人是掩码机制的示意性示例。

方向、夹爪开度、碰撞约束和机器人运动学由下游操作策略求解。

物理感知规划数据。对于操作规划，RynnBrain 1.1 遵循 RynnBrain [20]，的结构化规划格式，其中高层任务指令与涉及物体、区域和可供性的具身子任务描述配对。规划数据混合集包括 AgibotWorld [18]，Open X-Embodiment [16]，RynnBrain-Planning，以及新增的 Galaxea-G0。对于现有的具身规划数据，我们沿用与 RynnBrain 1.0 相同的标注格式和转换协议。对于 Galaxea-G0，原始标注提供子任务级标签，但不包含明确的高层任务指令。因此，我们以标注 ID 和已标注子任务的序列为条件，使用 Gemini 3.1 Pro 构建相应的全局任务描述。随后，将所得数据转换为与规划混合集其余部分相同的具身规划格式。

4 视觉-语言-动作模型的后训练

我们将下游 VLA 后训练既作为一种实际能力，也作为检验基础模型质量的一种探针。本节介绍 VLA 方法：策略架构（第 4.1 节）、统一动作空间设计（第 4.2 节）、跨具身/跨驱动器的部署框架（第 4.3 节）、用于平滑低延迟控制的实时分块（Real-Time Chunking）（第 4.4 节），以及所有骨干网络共享的微调设置（第 4.5 节）。相应的真实机器人结果在第 5.6 节报告。

4.1 模型架构

如图3所示，RynnBrain-1.1 中 RynnBrain-VLA 的模型架构大体沿用 RynnBrain-1.0 [20]，中的架构，其中 VLM 骨干本身作为单流扩散 Transformer，并在每一层处理视觉、语言和动作之间的密集交互。采用流匹配框架来预测每个步骤的动作块 [5]。为保留 VLM 固有的指令遵循能力，我们使用其原生对话格式来组织输入：

```
<|im_start|>user
INSTRUCTION:
Pour the wine from the bottle into the wine glass and place the glass on the table.
OBSERVATION:
<camera_1><camera_2><camera_3>
STATE:
<state>
What action should the robot take ?<|im_end|>
<|im_start|>assistant
<action>
```

Following π_0 [5], actions are positioned at the end of the sequence to enable the KV cache during inference.

4.2 Unified Cross-Embodiment Action Space

As illustrated in Figure 3, RynnBrain-VLA represents actions of heterogeneous robot embodiments within a shared high-dimensional action space. Before training, inspired by [41], we partition the unified action vector into semantically aligned groups corresponding to different body parts and control modes, including left and right arms, parallel grippers, dexterous hands, torso, and head control. Each embodiment activates only the action groups that are physically available on that robot, while all remaining dimensions are suppressed by an embodiment-specific mask.

The action loss is computed only on the active dimensions. For the Unitree G1 humanoid, the policy activates 14 dimensions (out of 40D) of the Hand group within the unified action space. These 14 dimensions correspond to the three-fingered dexterous hands of G1 with 7D per hand. In addition, the policy predicts a separate 64-dimensional SONIC [42] latent motion tokens that are not part of the unified action space. The 14-dimensional hand prediction and the 64-dimensional SONIC token are concatenated into a 78-dimensional representation and then fed to the SONIC whole-body controller to calculate target joint positions for the Unitree G1. For Tianji-Wuji, the policy activates the Arm-Joint group (14D, 7D per arm) and the Hand group (40D, 20D per hand), resulting in 54 active action dimensions. Astribot activates the Arm-Joint (14D, two arms), Gripper (2D), Head (3D), and Torso (4D), while masking the dexterous-hand joints.

This formulation allows trajectories from different robots to be mixed within the same training batch and optimized by a single VLA policy. The shared action components, such as left- and right-arm joint motion, receive supervision from multiple embodiments, whereas the embodiment-specific components are updated only by the corresponding data. Consequently, robot-independent manipulation features—including goal-directed arm motion, grasping, bimanual coordination, and task-level interaction patterns—can be shared across embodiments without requiring explicit alignment of their heterogeneous low-level action spaces. The unified action space not only serves as a common output interface, but also provides a mechanism for cross-embodiment knowledge sharing while preserving embodiment-specific control.

4.3 Cross-Embodiment and Cross-Driver Deployment

Real-world deployment requires a single policy interface to drive robots with very different configurations and low-level controllers. We therefore design a cross-embodiment, cross-driver framework that decouples the VLA policy from embodiment-specific execution, enabling the *same* RynnBrain-VLA recipe to be adapted to heterogeneous robots. The framework has three components:

- **Model layer.** Runs the VLA policy to produce a 32-step action chunk in the standard action space.
- **Control layer.** Two independent loops: a low-frequency loop (30 Hz) coordinating inference timing, the user interface, and action source switching; and a high-frequency loop (200 Hz) that interpolates the model's action chunk to the target control frequency and sends commands to the robot. The two loops communicate via shared memory.
- **Embodiment layer.** Dispatches standard-format actions (left/right arm, gripper, hand, torso) to the corresponding robot actuators and formats the robot's current state back for the model layer. Multi-

```
<|im_start|>user 指令：将酒从瓶子倒入酒杯，并将杯子放在桌上。观察：<
camera 1><camera 2><camera 3>状态：<
state>机器人应该采取什么动作？<|im_end|><|im_start|>assistant<action>
```

遵循 π_0 [5]，将动作置于序列末尾，以便在推理期间启用 KV 缓存。

4.2 统一的跨具身动作空间

如图 3 所示，RynnBrain-VLA 在共享的高维动作空间中表示异构机器人具身的动作。在训练之前，受 [41]，启发，我们将统一动作向量划分为语义对齐的若干组，这些组分别对应不同的身体部位和控制模式，包括左臂和右臂、平行夹爪、灵巧手、躯干以及头部控制。每个具身仅激活该机器人物理上可用的动作组，而所有其余维度则由具身特定的掩码抑制。

动作损失仅在激活维度上计算。对于 Unitree G1 人形机器人，策略会激活统一动作空间中 Hand 组的 14 个维度（共 40D）。这 14 个维度对应 G1 的三指灵巧手，每只手 7D。此外，策略还预测单独的 64 维 SONIC [42] 潜在运动 token，它们不属于统一动作空间。14 维手部预测和 64 维 SONIC token 会拼接为 78 维表示，然后输入 SONIC 全身控制器，以计算 Unitree G1 的目标关节位置。对于 Tianji-Wuji，策略会激活 Arm-Joint 组（14D，每条手臂 7D）和 Hand 组（40D，每只手 20D），从而得到 54 个激活动作维度。Astribot 会激活 Arm-Joint（14D，双臂）、Gripper（2D）、Head（3D）和 Torso（4D），同时掩蔽灵巧手关节。

这种构造方式允许将来自不同机器人的轨迹混合在同一训练批次中，并由单个 VLA 策略进行优化。共享动作分量（例如左臂和右臂关节运动）接受来自多个具身的监督，而具身特定分量仅由对应数据更新。因此，与机器人无关的操作特征——包括目标导向的手臂运动、抓取、双臂协调和任务级交互模式——可以在不同具身之间共享，而无需显式对齐其异构的底层动作空间。统一动作空间不仅作为公共输出接口，还提供了一种跨具身知识共享机制，同时保留具身特定的控制。

4.3 跨具身与跨驱动部署

真实世界部署需要单一的策略接口来驱动配置和底层控制器差异很大的机器人。因此，我们设计了一个跨具身、跨驱动的框架，将 VLA 策略与具身特定执行解耦，使相同的 RynnBrain-VLA 方案能够适配异构机器人。该框架包含三个组件：

- 模型层。运行 VLA 策略，在标准动作空间中生成 32 步动作块。
- 控制层。两个独立循环：一个低频循环（30 Hz），负责协调推理时序、用户界面和动作源切换；另一个高频循环（200 Hz），将模型的动作块插值到目标控制频率，并向机器人发送指令。两个循环通过共享内存通信。
- 具身层负责将标准格式的动作（左/右臂、夹爪、手、躯干）分发给对应的机器人执行器，并将机器人的当前状态格式化后回传给模型层。多

camera images are packaged as a dictionary whose keys match the naming convention used during training.

This design allows adding a new embodiment by only implementing the embodiment layer, without modifying the model or control layers. We instantiate this framework on three heterogeneous embodiments—the Unitree G1 humanoid, the Astribot-S1 bimanual manipulator, and the Tianji-Wuji dexterous-hand system—covering distinct action spaces, degrees of freedom, and end-effectors.

4.4 Real-Time Chunking (RTC)

We adopt **Real-Time Chunking (RTC)** [7]. The model predicts a 32-step action chunk but does not wait for the entire chunk to be executed: a new inference is triggered every 5 steps while the current chunk is still running. The unexecuted portion of the previous chunk is then used to guide the denoising of the new chunk, following the action-guidance formulation of Black et al. [7] (Sec. 3.1) with strength $\beta = 10.0$. Each step in the previous chunk is assigned a guidance weight reflecting whether it will be consumed before the new chunk arrives: (i) the initial 5 steps and the steps consumed during inference (estimated as the sliding average of the past ten inference times) receive weight $= 1$, as they are guaranteed to be executed before the new chunk is ready; (ii) a transition region where the weight decreases smoothly from 1 to 0; and (iii) the tail of the chunk, which receives weight $= 0$ with no consistency enforced. This ensures smooth chunk-boundary continuity while allowing the model to freely adapt later predictions to new observations.

4.5 Fine-tuning Setup

We collect teleoperation demonstrations on the three target embodiments described above, covering a range of pick-and-place and manipulation tasks with different objects and interaction requirements. Each demonstration contains synchronized visual observations, robot states, and action trajectories, together with a task instruction specifying the target object or placement location.

For the controlled comparison between the Qwen-based VLA and RynnBrain-VLA in Section 5.6, we use the same demonstration data, VLA post-training pipeline, and optimization settings. Both policies are fine-tuned for 60k steps with a learning rate of 2×10^{-5} and a batch size of 32. All input images are proportionally resized such that the short side is 384 pixels. This setup reduces differences introduced by downstream training and allows us to examine how the choice of base model affects the resulting real-robot policy. The fine-tuned policies are evaluated quantitatively in Section 5.6.

5 Evaluation

We organize our evaluation into several parts. We first assess the base models on embodied cognition and localization benchmarks across three model scales. At the 2B scale, we compare RynnBrain 1.1 with RynnBrain 1.0, Cosmos 3-Edge [49], SenseNova-SI-1.1-InternVL3-2B [11], Cosmos-Reason2-2B [2], and Qwen3.5-2B [53]. At the 4B–9B scale, we compare against RynnBrain 1.0, Cosmos 3-Nano [49], RoboBrain-2.5-4B [62], Thinker-4B [51], Molmo2-ER-5B [15], Pelican-VL-7B [77], MiMo-Embodied-7B [28], Cosmos-Reason2-8B [2], SenseNova-SI1.5-8B [11], and Qwen3.5 [53]. At the largest scale, we compare with Cosmos 3-Super, RoboBrain 2.0-32B, Cosmos-Reason2-32B, Pelican-VL-72B [77], Qwen3.5-122B-A10B [53], Gemini 3 Pro [63], GPT 5.4 [50], Claude Sonnet 4.6 [1], Gemini Robotics-ER 1.5 [64], and HY-Embodied 0.5 [65]. We then analyze scaling against the corresponding Qwen3.5 models, evaluate native 3D grounding and contact-point prediction, and finally examine RynnBrain-VLA through controlled real-robot evaluations across three embodiments

5.1 Embodied Cognition Capability

We evaluate RynnBrain 1.1 on a diverse suite of embodied cognition benchmarks covering video-based spatial intelligence, multi-view reasoning, embodied question answering, robot-oriented spatial understanding, and object-centric cognition. The evaluation includes VSI-Bench [68], MMSI [70], ERQA [64], RoboSpatial-VQA [60], MindCube [71], EmbSpatial [21], RynnBrain-Object, and RynnBrain-Spatial. As shown in Tables 3

相机图像被打包为一个字典，其键与训练时使用的命名约定相匹配。

该设计允许仅通过实现具身层来添加新的具身平台，而无需修改模型层或控制层。我们在三个异构具身平台上实例化了该框架——Unitree G1 人形机器人、Astribot-S1 双臂机械臂和 Tianji-Wuji 灵巧手系统——它们覆盖不同的动作空间、自由度和末端执行器。

4.4 实时分块（RTC）

我们采用实时分块（Real-Time Chunking，RTC）[7]。模型预测一个 32 步动作块，但并不等待整个动作块执行完毕：在当前动作块仍在执行时，每 5 步触发一次新的推理。随后，前一个动作块中尚未执行的部分被用于引导新动作块的去噪，遵循 Black 等人 [7]（第 3.1 节）的动作引导公式，强度为 $\beta = 10.0$。前一个动作块中的每一步都会被分配一个引导权重，以反映它是否会在新动作块到达前被执行：(i) 最初的 5 步以及推理期间消耗的步骤（估计为过去十次推理时间的滑动平均值）获得权重 $= 1$，因为可以保证它们会在新动作块就绪前执行；(ii) 权重从 1 平滑降至 0 的过渡区域；以及 (iii) 动作块尾部，其权重为 $= 0$，不强制一致性。这确保了动作块边界的平滑连续性，同时允许模型自由地根据新观测调整后续预测。

4.5 微调设置

我们在上述三种目标本体上采集遥操作示范，涵盖一系列拾取放置与操作任务，涉及不同物体和交互需求。每个示范包含同步的视觉观测、机器人状态和动作轨迹，以及指定目标物体或放置位置的任务指令。

对于第 5.6 节中基于 Qwen 的 VLA 与 RynnBrain-VLA 之间的受控比较，我们使用相同的示范数据、VLA 后训练流程和优化设置。两个策略均微调 60k 步，学习率为 2×10^{-5}，批量大小为 32。所有输入图像均按比例缩放，使短边为 384 像素。该设置减少了由下游训练引入的差异，并使我们能够考察基础模型的选择如何影响最终的真实机器人策略。微调后的策略将在第 5.6 节进行定量评估。

5 评估

我们将评估组织为若干部分。我们首先在三种模型规模上，基于具身认知与定位基准对基础模型进行评估。在 2B 规模下，我们将 RynnBrain 1.1 与 RynnBrain 1.0、Cosmos 3-Edge [49], SenseNova-SI-1.1-InternVL3-2B [11], Cosmos-Reason2-2B [2], 和 Qwen3.5-2B [53] 进行比较。在 4B–9B 规模下，我们将与 RynnBrain 1.0、Cosmos 3-Nano [49], RoboBrain-2.5-4B [62], Thinker- 4B [51], Molmo2-ER-5B [15], Pelican-VL-7B [77], MiMo-Embodied-7B [28], Cosmos-Reason2-8B [2], SenseNova- SI1.5-8B [11], 和 Qwen3.5 [53] 进行比较。在最大规模下，我们将与 Cosmos 3-Super、RoboBrain 2.0- 32B、Cosmos-Reason2-32B、Pelican-VL-72B [77], Qwen3.5-122B-A10B [53], Gemini 3 Pro [63], GPT 5.4 [50],Claude Sonnet 4.6 [1], Gemini Robotics-ER 1.5 [64], 和 HY-Embodied 0.5 [65] 进行比较。随后，我们分析相对于对应 Qwen3.5 模型的扩展性，评估原生 3D 定位与接触点预测，最后通过跨三种具身形态的受控真实机器人评估来考察 RynnBrain-VLA

5.1 具身认知能力

我们在涵盖基于视频的空间智能、多视角推理、具身问答、面向机器人的空间理解以及以对象为中心的认知等多样化的具身认知基准上评估 RynnBrain 1.1。评估包括 VSI-Bench [68], MMSI [70], ERQA [64], RoboSpatial- VQA [60], MindCube [71], EmbSpatial [21], RynnBrain-Object 和 RynnBrain-Spatial。如表 3

Table 3 Comparison with lightweight (2B-scale) models on embodied cognition and localization benchmarks. * denotes results from our reproduction.

Benchmark	RynnBrain 1.1 2B	RynnBrain 2B	Cosmos 3 -Edge 2B	SenseNova-SI-1.1 InternVL3 2B	Cosmos -Reason2 2B	Qwen3.5 2B
VSI-Bench	**72.9**	70.5	59.2	63.7*	46.3*	44.1
MMSI	**40.5**	34.1	32.3	34.2	27.8*	28.3
ERQA	**42.3**	42.3	42.0	28.7*	37.8	33.0
RoboSpatial-Pointing	59.6	**65.6**	-	0.0*	11.5*	-
RoboSpatial-VQA	60.4	64.9	-	24.8*	**65.3***	-
RoboSpatial-Overall	62.1	**65.2**	-	16.1*	47.4*	41.5
MindCube	**61.7**	50.1	-	41.8	33.7*	37.5
EmbSpatial	73.9	**79.6**	-	63.0*	65.6*	67.7*
RynnBrain-Object	**74.6**	70.7	24.0	20.7	25.8	37.1*
RynnBrain-Spatial	**63.8**	57.2	22.6	13.0	24.3	15.6*
RefSpatial-Bench	**58.5**	52.7	48.4	5.6*	23.6*	30.0
RynnBrain-Grounding	**84.1**	79.1	51.6	12.6	14.5	24.2*
RynnBrain-Area	**57.2**	54.6	39.1	8.6	22.1	8.1*
RynnBrain-Affordance	**90.2**	89.4	77.6	57.8	55.5	80.4*
RynnBrain-Trajectory	**68.8**	66.6	58.7	49.0	43.7	28.3*

and 4, RynnBrain 1.1 improves substantially over the previous RynnBrain models on reasoning-intensive and spatially demanding tasks. At the 2B scale, the improvements are task-dependent: MMSI increases from 34.1 to 40.5, MindCube from 50.1 to 61.7, and RynnBrain-Spatial from 57.2 to 63.8, while several RoboSpatial and EmbSpatial results remain below RynnBrain-2B. The gains become more systematic at the 9B scale, where RynnBrain 1.1 outperforms RynnBrain-8B on eight of ten shared benchmarks. In particular, MindCube improves from 56.6 to 86.9, MMSI from 39.6 to 47.0, and RynnBrain-Spatial from 59.9 to 67.9, indicating stronger multi-view and spatial reasoning. Compared with other models at similar scales, RynnBrain 1.1 achieves leading results on multiple cognition benchmarks at both the 2B and 9B scales. RynnBrain 1.1-122B-A10B further extends this advantage to the large-model regime, with performance increasing from 40.5 to 47.0 and 52.0 on MMSI, and from 61.7 to 86.9 and 89.6 on MindCube, when scaling from 2B to 9B and 122B-A10B. These results show that model scaling is particularly beneficial for reasoning-intensive embodied cognition.

5.2 Embodied Localization Capability

We evaluate embodied localization across six spatial grounding tasks: RefSpatial-Bench [80],RoboSpatial-Pointing [60], RynnBrain-Grounding, RynnBrain-Area, RynnBrain-Affordance, and RynnBrain-Trajectory. These benchmarks cover language-guided spatial reference, object and region localization, affordance grounding, and trajectory prediction. RynnBrain 1.1 improves over the previous RynnBrain models on location benchmarks at both the 2B and 9B scales. At 2B, RefSpatial-Bench increases from 52.7 to 58.5 and RynnBrain-Grounding from 79.1 to 84.1. At 9B, the gains become larger on relational and motion-oriented tasks: RefSpatial-Bench improves from 59.2 to 67.2, while RynnBrain-Trajectory increases from 64.5 to 70.0. The relatively small gain on RynnBrain-Affordance reflects its already high performance in RynnBrain 1.0. Both RynnBrain 1.1-2B and RynnBrain 1.1-9B achieve the best results among models in their respective scale groups on most of these location benchmarks. At the large scale, RynnBrain 1.1-122B-A10B achieves leading performance on RefSpatial-Bench, RynnBrain-Grounding, and RynnBrain-Affordance. The clearest scaling trend appears on RefSpatial-Bench, which improves from 58.5 at 2B to 67.2 at 9B and 79.1 at 122B-A10B, showing that relational spatial grounding benefits strongly from increased model capacity.

表 3 比较 wi th 轻量级（2B 规模）模型在具身认知和定位基准上的表现。* 表示 s 来自我们 repro 的结果 duction。

基准	RynnBrain 1.1 2B	RynnBrain 2B	Cosmos 3 -Edge 2B	SenseNova-SI-1.1 InternVL3 2B	Cosmos -Reason2 2B	Qwen3.5 2B
VSI-Bench	**72.9**	70.5	59.2	63.7*	46.3*	44.1
MMSI	**40.5**	34.1	32.3	34.2	27.8*	28.3
ERQA	**42.3**	42.3	42.0	28.7*	37.8	33.0
RoboSpatial-Pointing	59.6	**65.6**	-	0.0*	11.5*	-
RoboSpatial-VQA	60.4	64.9	-	24.8*	**65.3***	-
RoboSpatial-总体	62.1	**65.2**	-	16.1*	47.4*	41.5
MindCube	**61.7**	50.1	-	41.8	33.7*	37.5
EmbSpatial	73.9	**79.6**	-	63.0*	65.6*	67.7*
RynnBrain-Object	**74.6**	70.7	24.0	20.7	25.8	37.1*
RynnBrain-Spatial	**63.8**	57.2	22.6	13.0	24.3	15.6*
RefSpatial-Bench	**58.5**	52.7	48.4	5.6*	23.6*	30.0
RynnBrain-Grounding	**84.1**	79.1	51.6	12.6	14.5	24.2*
RynnBrain-Area	**57.2**	54.6	39.1	8.6	22.1	8.1*
RynnBrain-Affordance	**90.2**	89.4	77.6	57.8	55.5	80.4*
RynnBrain-Trajectory	**68.8**	66.6	58.7	49.0	43.7	28.3*

和 4 所示，RynnBrain 1.1 在推理密集型与空间要求高的任务上相较此前的 RynnBrain 模型取得了显著提升。在 2B 规模下，提升幅度因任务而异：MMSI 从 34.1 提升至 40.5，MindCube 从 50.1 提升至 61.7，RynnBrain-Spatial 从 57.2 提升至 63.8，而若干 RoboSpatial 和 EmbSpatial 结果仍低于 RynnBrain-2B。在 9B 规模下，这种增益变得更加系统性：RynnBrain 1.1 在十个共有基准中的八个上优于 RynnBrain-8B。具体而言，MindCube 从 56.6 提升至 86.9，MMSI 从 39.6 提升至 47.0，RynnBrain-Spatial 从 59.9 提升至 67.9，表明其多视角与空间推理能力更强。与相近规模的其他模型相比，RynnBrain 1.1 在 2B 和 9B 规模下均在多项认知基准上取得了领先结果。RynnBrain 1.1- 122B-A10B 进一步将这一优势扩展至大模型区间：当从 2B 扩展至 9B 和 122B-A10B 时，MMSI 上的性能从 40.5 提升至 47.0 和 52.0，MindCube 上从 61.7 提升至 86.9 和 89.6。这些结果表明，模型扩展对推理密集型具身认知尤为有益。

5.2 具身定位能力

我们在六个空间定位任务上评估具身定位能力：RefSpatial-Bench [80],RoboSpatial- Pointing [60], RynnBrain-Grounding、RynnBrain-Area、RynnBrain-Affordance 和 RynnBrain-Trajectory。这些基准涵盖语言引导的空间参照、物体与区域定位、可供性定位以及轨迹预测。RynnBrain 1.1 在 2B 和 9B 规模上均较之前的 RynnBrain 模型在定位基准上有所提升。在 2B 规模下，RefSpatial-Bench 从 52.7 提升至 58.5，RynnBrain-Grounding 从 79.1 提升至 84.1。在 9B 规模下，关系型和运动导向任务上的增益更大：RefSpatial-Bench 从 59.2 提升至 67.2，而 RynnBrain-Trajectory 从 64.5 提升至 70.0。RynnBrain-Affordance 上的增益相对较小，反映出其在 RynnBrain 1.0 中已具有较高性能。在大多数这些定位基准上，RynnBrain 1.1-2B 和 RynnBrain 1.1-9B 均在其各自规模组内的模型中取得了最佳结果。在大规模方面，RynnBrain 1.1-122B-A10B 在 RefSpatial-Bench、RynnBrain-Grounding 和 RynnBrain-Affordance 上取得了领先性能。最明显的规模扩展趋势出现在 RefSpatial-Bench 上：其性能从 2B 的 58.5 提升到 9B 的 67.2，再到 122B-A10B 的 79.1，表明关系型空间定位会从模型容量增加中显著受益。

Table 4 Comparison with 4B–9B scale models on embodied cognition and localization benchmarks. * denotes results from our reproduction.

Benchmark	RynnBrain 1.1 9B	RynnBrain 8B	Cosmos 3 -Nano 8B	RoboBrain-2.5 4B	Thinker 4B	Molmo2-ER 5B	Pelican-VL 7B	MiMo-Embodied 7B	Cosmos -Reason2 8B	SenseNova -SI1.5 8B	Qwen3.5 9B
VSI-Bench	**74.9**	70.9	54.9	41.7	65.4	74.5	52.8	48.5	53.7*	67.3*	61.0
MMSI	**47.0**	39.6	36.2	20.5	27.7*	43.8	26.2*	30.2*	31.3*	38.3*	13.4
ERQA	**47.5**	46.8	46.0	43.3	42.5*	46.8	39.8*	46.8	46.0*	40.0*	41.5*
RoboSpatial-Pointing	**69.7**	60.1	33.0*	-	-	32.0	-	-	24.6	29.3*	48.7
RoboSpatial-VQA	68.0	**80.0**	77.6*	-	-	73.4	-	-	76.3	61.4*	57.0
RoboSpatial-Overall	69.1	**73.1**	62.1*	62.3	70.8	59.0*	57.5	61.8	59.0*	40.1*	54.1*
MindCube	86.9	56.6	40.5*	26.9	39.0*	57.0	33.7*	43.1*	43.8*	**92.1**	33.1
EmbSpatial	81.9	80.4	81.4*	76.9	80.2	78.8	76.6	76.2	77.0*	80.3	**83.0**
RynnBrain-Object	**75.9**	71.2	39.8	36.4	26.5	17.1	30.8	39.0	37.2	16.8	45.6
RynnBrain-Spatial	**67.9**	59.9	26.1	32.6	17.3	19.7	20.5	28.3	31.4	19.9	28.4
RefSpatial-Bench	**67.2**	59.2	53.1	56.0	61.0	52.5	22.3	48.0	33.1*	8.6*	37.3
RynnBrain-Grounding	**85.4**	81.6	72.9	5.4	2.4	4.2	3.5	49.8	60.0	44.7	33.0
RynnBrain-Area	**59.6**	56.2	52.0	42.4	33.8	30.6	46.5	49.4	37.6	11.3	32.9
RynnBrain-Affordance	**90.7**	90.4	85.1	69.1	59.9	81.7	81.4	84.4	83.9	64.9	60.5
RynnBrain-Trajectory	**70.0**	64.5	67.9	63.0	49.2	59.3	59.2	61.3	64.0	53.9	45.0

5.3 Scaling Analysis

A central contribution of RynnBrain 1.1 is studying how embodied capability evolves with model scale under a unified training recipe. We compare RynnBrain 1.1 (2B, 9B, 122B-A10B) against the corresponding raw Qwen3.5 baselines at matched model sizes, grouping benchmarks into three categories: *general embodied cognition* (visual recognition, spatial understanding, and object-centric reasoning), *reasoning-intensive cognition* (multi-view spatial reasoning and video-based temporal inference), and *embodied localization* (language-guided spatial grounding and trajectory prediction). Figure 4 summarizes the scaling trends across these groups.

Embodied capabilities exhibit non-uniform scaling laws. Unlike standard vision–language benchmarks where scaling consistently improves performance, we observe three distinct scaling regimes in the embodied domain. (1) For *general embodied cognition*, both RynnBrain 1.1 and Qwen3.5 improve monotonically with scale, and the performance gap steadily narrows. This mirrors conventional MLLM scaling—larger models reliably improve on recognition and comprehension tasks, and general model scaling can partially compensate for the lack of embodied supervision. (2) For *reasoning-intensive cognition*, the two models diverge fundamentally: RynnBrain 1.1 improves steadily (+38.6%), while Qwen3.5 exhibits *negative scaling* (−39.2%), with the gap widening from 18.2 to 50.8. This indicates that multi-view and temporal reasoning capabilities are not emergent from general VLM scaling but require explicit embodied supervision as a prerequisite. Without spatiotemporal grounding, stronger language priors in larger models may override weak visual-spatial signals, producing more confident but less accurate responses. (3) For *embodied localization*, both models improve substantially, yet Qwen3.5 at its largest scale still underperforms RynnBrain 1.1 at its smallest. This suggests that for spatial-temporal pointing tasks, *data scaling*—training on explicit coordinate supervision—is more effective than model scaling alone. RynnBrain's embodied pretraining equips even its compact model with structured spatial output capabilities that pure language modeling must recover implicitly, requiring disproportionately more parameters to approximate.

Embodied pretraining as a scaling enabler. The decomposition in Figure 4(d) further illustrates this contrast. RynnBrain 1.1 consistently establishes a higher performance floor across all categories, reflecting the value of embodied data even at compact scales. More critically, on reasoning-intensive tasks, embodied pretraining determines whether scaling *helps or hurts*—it transforms a capability that degrades under pure VLM scaling into one that benefits consistently. These findings indicate that for the embodied domain, data-driven spatial supervision and model scaling are complementary rather than substitutable: scaling amplifies the benefit of embodied pretraining on reasoning-heavy tasks, while embodied supervision ensures that scaling produces

表4 比较 wi th 4B–9B 规模模型在具身认知与定位基准上的表现。* 表示结果 s 来自我们的复现 .

基准	RynnBrain 1.1 9B	RynnBrain 8B	Cosmos 3 -Nano 8B	RoboBrain-2.5 4B	Thinker 4B	Molmo2-ER 5B	Pelican-VL 7B	MiMo-具身 7B	Cosmos -Reason2 8B	SenseNova -SI1.5 8B	Qwen3.5 9B
VSI-Bench	**74.9**	70.9	54.9	41.7	65.4	74.5	52.8	48.5	53.7*	67.3*	61.0
MMSI	**47.0**	39.6	36.2	20.5	27.7*	43.8	26.2*	30.2*	31.3*	38.3*	13.4
ERQA	**47.5**	46.8	46.0	43.3	42.5*	46.8	39.8*	46.8	46.0*	40.0*	41.5*
RoboSpatial-Pointing	**69.7**	60.1	33.0*	-	-	32.0	-	-	24.6	29.3*	48.7
RoboSpatial-VQA	68.0	**80.0**	77.6*	-	-	73.4	-	-	76.3	61.4*	57.0
RoboSpatial-总体	69.1	**73.1**	62.1*	62.3	70.8	59.0*	57.5	61.8	59.0*	40.1*	54.1*
MindCube	86.9	56.6	40.5*	26.9	39.0*	57.0	33.7*	43.1*	43.8*	**92.1**	33.1
EmbSpatial	81.9	80.4	81.4*	76.9	80.2	78.8	76.6	76.2	77.0*	80.3	**83.0**
RynnBrain-Object	**75.9**	71.2	39.8	36.4	26.5	17.1	30.8	39.0	37.2	16.8	45.6
RynnBrain-Spatial	**67.9**	59.9	26.1	32.6	17.3	19.7	20.5	28.3	31.4	19.9	28.4
RefSpatial-Bench	**67.2**	59.2	53.1	56.0	61.0	52.5	22.3	48.0	33.1*	8.6*	37.3
RynnBrain-Grounding	**85.4**	81.6	72.9	5.4	2.4	4.2	3.5	49.8	60.0	44.7	33.0
RynnBrain-Area	**59.6**	56.2	52.0	42.4	33.8	30.6	46.5	49.4	37.6	11.3	32.9
RynnBrain-Affordance	**90.7**	90.4	85.1	69.1	59.9	81.7	81.4	84.4	83.9	64.9	60.5
RynnBrain-Trajectory	**70.0**	64.5	67.9	63.0	49.2	59.3	59.2	61.3	64.0	53.9	45.0

5.3 缩放分析

RynnBrain 1.1 的一项核心贡献是研究在统一训练配方下，具身能力如何随模型规模演变。我们在相同模型规模下，将 RynnBrain 1.1（2B、9B、122B-A10B）与相应的原始 Qwen3.5 基线进行比较，并将基准分为三类：通用具身认知（视觉识别、空间理解和以物体为中心的推理）、推理密集型认知（多视角空间推理和基于视频的时序推理）以及具身定位（语言引导的空间定位和轨迹预测）。图 4 总结了这些组别之间的规模扩展趋势。

具身能力呈现出非均匀的缩放规律。与标准视觉—语言基准中缩放能持续提升性能不同，我们在具身领域观察到三种不同的缩放模式。（1）对于通用具身认知，RynnBrain 1.1 与 Qwen3.5 都随规模增大而单调提升，且性能差距稳步缩小。这与传统 MLLM 缩放规律一致——更大的模型能在识别与理解任务上可靠提升，而通用模型缩放可部分弥补具身监督的不足。（2）对于推理密集型认知，两个模型出现根本分化：RynnBrain 1.1 稳步提升（+38.6%），而 Qwen3.5 出现负缩放（−39.2%），差距从 18.2 扩大到 50.8。这表明多视图和时序推理能力并非从通用 VLM 缩放中涌现，而是需要显式具身监督作为前提。在缺乏时空定位的情况下，更大模型中更强的语言先验可能会压倒较弱的视觉—空间信号，从而产生更自信但更不准确的回答。（3）对于具身定位，两个模型均有显著提升，但 Qwen3.5 在其最大规模下仍不如 RynnBrain 1.1 在其最小规模下的表现。这表明对于时空指向任务，数据缩放——在显式坐标监督上训练——比单纯的模型缩放更有效。RynnBrain 的具身预训练使其即使紧凑模型也具备结构化空间输出能力，而纯语言建模必须隐式地恢复这种能力，并且需要不成比例地更多参数才能近似达到。

具身预训练作为扩展的赋能因素。图4(d)中的分解进一步说明了这一对比。RynnBrain 1.1 在所有类别中始终建立起更高的性能下限，反映出即便在较小规模下具身数据的价值。更关键的是，在推理密集型任务上，具身预训练决定扩展是有益还是有害——它将一种在纯 VLM 扩展下会退化的能力，转变为能够持续受益的能力。这些发现表明，在具身领域，数据驱动的空间监督与模型扩展是互补而非可替代的：扩展放大了具身预训练在推理密集任务上的收益，而具身监督确保扩展产生

Table 5 Comparison with large-scale open-source and proprietary models on embodied cognition and localization benchmarks. * denotes results from our reproduction.

Benchmark	RynnBrain 1.1 122B-A10B	Cosmos 3 Super 32B	RoboBrain 2.0 32B	Cosmos-Reason2 32B	Pelican-VL 72B	Qwen3.5 122B-A10B	Gemini 3 Pro -	GPT 5.4 -	Claude Sonnet 4.6 -	Gemini Robotics -ER 1.5 -	HY-Embodied 0.5 A32B
VSI-Bench	**75.0**	60.9	42.7	67.0*	57.3	66.6*	48.8*	49.2*	44.4*	45.8	68.3
MMSI	**52.0**	41.8	28.5*	32.1*	30.7*	9.2*	49.2	37.5*	33.0*	-	39.2
ERQA	54.3	51.2	46.0	45.3	43.0	62.0	**70.5**	47.5*	44.8*	54.8	62.3
RoboSpatial-Pointing	**69.7**	24.2*	-	34.4	-	59.8	27.0*	20.8*	5.8*	31.1	-
RoboSpatial-VQA	70.6	74.1*	-	67.1	-	68.9	66.7*	75.0*	55.3*	**79.3**	-
RoboSpatial-Overall	70.3	56.7*	**72.4**	55.4	55.4	65.7*	53.0*	56.1*	38.0*	62.6	-
MindCube	**89.6**	40.2*	29.2	39.4*	32.5*	30.8*	70.8	10.4*	21.0*	54.7	69.2
EmbSpatial	83.7	**83.8***	73.8*	78.5*	76.6	80.5	83.6	81.8*	61.5*	78.4	-
RynnBrain-Object	**77.0**	48.3	26.2	36.8*	42.2	55.4	58.4	55.5	33.5	-	-
RynnBrain-Spatial	**70.0**	34.5	11.6	38.4*	32.2	31.0	38.5	43.3	36.7	-	-
RefSpatial-Bench	**79.1**	57.0	54.0	45.5*	49.5	69.3	65.5	26.7*	14.6*	48.5	57.2
RynnBrain-Grounding	**86.6**	74.4	0.0	60.0	10.8	66.6	62.1	24.2	3.3	-	-
RynnBrain-Area	60.7	53.0	45.3	37.6	53.2	41.9	**61.5**	37.1	15.8	-	-
RynnBrain-Affordance	**91.3**	84.6	76.1	83.9	87.3	79.0	86.0	83.4	60.0	-	-
RynnBrain-Trajectory	69.7	69.3	60.3	64.0	64.1	62.8	72.0	**72.8**	56.7	-	-

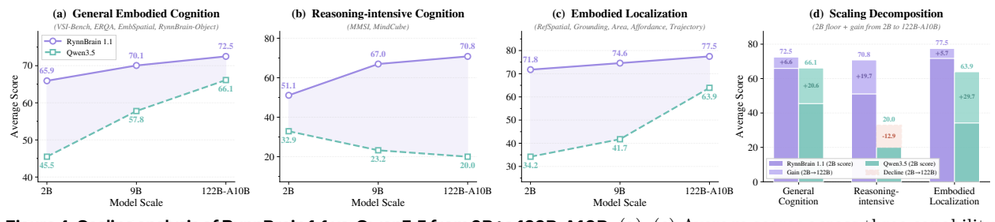

Figure 4 Scaling analysis of RynnBrain 1.1 vs. Qwen3.5 from 2B to 122B-A10B. (a)–(c) Average scores across three capability groups: general embodied cognition, reasoning-intensive cognition, and embodied localization. (d) Scaling decomposition showing the 2B performance floor and the gain from scaling to 122B-A10B.

reliable gains rather than regression.

5.4 3D Grounding

We evaluate the native 3D grounding capability of RynnBrain 1.1 at both the 2B and 9B scales. On SUN RGB-D, RynnBrain 1.1-2B achieves 34.28 AP@15, already outperforming general-purpose VLMs such as Seed1.5-VL (33.5) and Gemini 2.0 Pro (32.5). Scaling the model to 9B further improves the result to **41.12 AP@15**, substantially narrowing the gap to the closed-source Gemini Robotics-ER (48.3). These results show that explicit 3D-grounded training can equip even compact models with metric spatial understanding, while larger model capacity further strengthens depth, size, and orientation reasoning.

A similar scaling trend is observed on WildDet3D-Bench. RynnBrain 1.1-2B obtains 17.36 AP3D, while RynnBrain 1.1-9B reaches **23.44 AP3D**, surpassing the specialized WildDet3D detector trained with additional in-domain data (22.6). The improvement from 2B to 9B indicates that native 3D grounding benefits from model scaling in addition to explicit 3D supervision. Overall, RynnBrain 1.1 demonstrates strong language-conditioned 3D grounding across model sizes, with the 9B model achieving competitive performance against both specialized detectors and substantially larger proprietary systems.

In addition to the quantitative evaluation, we visualize representative predictions in Figure 6. RynnBrain 1.1 can localize target objects and estimate their spatial extent and orientation from a single image, including large furniture such as beds, tables, and chairs. These examples illustrate the model's ability to produce physically grounded 3D predictions across diverse viewpoints and scene layouts.

表 5 对比与大规模开源和专有模型在具身认知与定位基准测试。* 表示结果来自我们的复现。

基准测试	RynnBrain 1.1 122B-A10B	Cosmos 3 Super 32B	RoboBrain 2.0 32B	Cosmos-Reason2 32B	Pelican-VL 72B	Qwen3.5 122B-A10B	Gemini 3 Pro -	GPT 5.4 -	Claude Sonnet 4.6 -	Gemini Robotics -ER 1.5 -	HY-Embodied 0.5 A32B
VSI-Bench	**75.0**	60.9	42.7	67.0*	57.3	66.6*	48.8*	49.2*	44.4*	45.8	68.3
MMSI	**52.0**	41.8	28.5*	32.1*	30.7*	9.2*	49.2	37.5*	33.0*	-	39.2
ERQA	54.3	51.2	46.0	45.3	43.0	62.0	**70.5**	47.5*	44.8*	54.8	62.3
RoboSpatial 指向	**69.7**	24.2*	-	34.4	-	59.8	27.0*	20.8*	5.8*	31.1	-
RoboSpatial-VQA	70.6	74.1*	-	67.1	-	68.9	66.7*	75.0*	55.3*	**79.3**	-
RoboSpatial-整体	70.3	56.7*	**72.4**	55.4	55.4	65.7*	53.0*	56.1*	38.0*	62.6	-
MindCube	**89.6**	40.2*	29.2	39.4*	32.5*	30.8*	70.8	10.4*	21.0*	54.7	69.2
EmbSpatial	83.7	**83.8***	73.8*	78.5*	76.6	80.5	83.6	81.8*	61.5*	78.4	-
RynnBrain-Object	**77.0**	48.3	26.2	36.8*	42.2	55.4	58.4	55.5	33.5	-	-
RynnBrain-Spatial	**70.0**	34.5	11.6	38.4*	32.2	31.0	38.5	43.3	36.7	-	-
RefSpatial-Bench	**79.1**	57.0	54.0	45.5*	49.5	69.3	65.5	26.7*	14.6*	48.5	57.2
RynnBrain-Grounding	**86.6**	74.4	0.0	60.0	10.8	66.6	62.1	24.2	3.3	-	-
RynnBrain-Area	60.7	53.0	45.3	37.6	53.2	41.9	**61.5**	37.1	15.8	-	-
RynnBrain-Affordance	**91.3**	84.6	76.1	83.9	87.3	79.0	86.0	83.4	60.0	-	-
RynnBrain-Trajectory	69.7	69.3	60.3	64.0	64.1	62.8	72.0	**72.8**	56.7	-	-

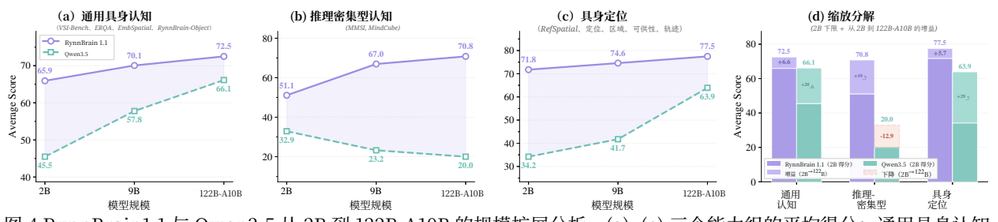

图 4 RynnBrain1.1 与 Qwen3.5 从 2B 到 122B-A10B 的规模扩展分析。(a)–(c) 三个能力组的平均得分：通用具身认知、推理密集型认知和具身定位。(d) 规模扩展分解，展示 2B 的性能下限以及扩展至 122B-A10B 所带来的增益。

可靠的收益而非回退。

5.4 3D 定位

我们在 2B 和 9B 两种规模下评估了 RynnBrain 1.1 的原生 3D 定位能力。在 SUN RGB-D 上，RynnBrain 1.1-2B 达到 34.28 AP@15，已经优于 Seed1.5-VL（33.5）和 Gemini 2.0 Pro（32.5）等通用 VLM。将模型扩展至 9B 后，结果进一步提升至 41.12 AP@15，显著缩小了与闭源 Gemini Robotics-ER（48.3）的差距。这些结果表明，显式的 3D 定位训练可使即使紧凑的模型也具备度量空间理解能力，而更大的模型容量进一步增强深度、尺寸和朝向推理能力。

在 WildDet3D-Bench 上也观察到类似的规模扩展趋势。RynnBrain 1.1-2B 获得 17.36 AP3D，而 RynnBrain 1.1-9B 达到 23.44 AP3D，超过了使用额外域内数据训练的专用 WildDet3D 检测器（22.6）。从 2B 到 9B 的提升表明，原生 3D 定位除了受益于显式 3D 监督外，还能从模型规模扩展中获益。总体而言，RynnBrain 1.1 在不同模型规模下均展现出强大的语言条件 3D 定位能力，其中 9B 模型在与专用检测器以及规模大得多的专有系统对比时，仍取得了有竞争力的性能。

除定量评估外，我们在图6中可视化了具有代表性的预测结果。RynnBrain 1.1 能够从单张图像中定位目标物体，并估计其空间范围和朝向，包括床、桌子和椅子等大型家具。这些示例展示了该模型能够在多样视角和场景布局下生成具有物理依据的3D预测。

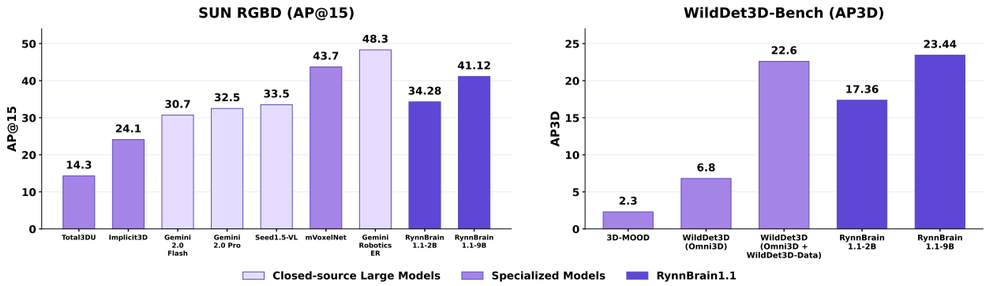

Figure 5 Performance Comparison on SUN RGB-D and WildDet3D-Bench with RynnBrain 1.1-2B and 9B.

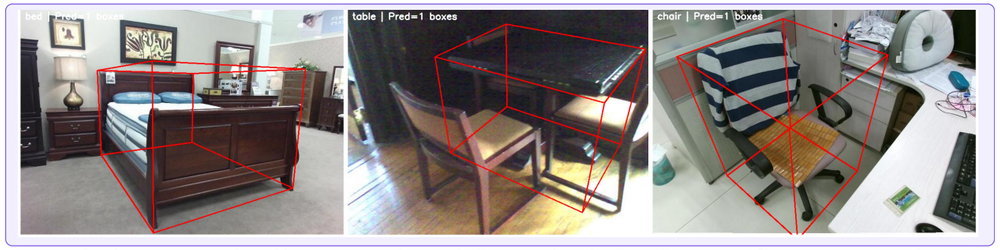

Figure 6 Qualitative examples of language-conditioned 3D grounding by RynnBrain 1.1. Given an image and a target object description, the model predicts an oriented 3D bounding box in the camera coordinate system. The examples cover objects with different scales, shapes, viewpoints, and levels of occlusion in indoor environments.

5.5 Contact Point Prediction

Evaluating contact point prediction remains challenging because there is currently no standardized metric that reliably reflects the functional validity of a predicted contact. Pixel-level distance to a single annotation may penalize alternative contact points that are equally valid for manipulation, while conventional grasp-box metrics are not directly applicable to our contact-centered representation. We therefore provide a qualitative evaluation in Figure 7 to illustrate the capability of RynnBrain 1.1 across diverse objects and instructions. Here we show the cases produced by RynnBrain 1.1-9B.

The visualized cases show that RynnBrain 1.1 can identify task-relevant contact locations rather than merely predicting object centers. For example, the model selects the handle when asked to pick up a mug, adapts its prediction when the mug is overturned, and produces appropriate contact locations and orientations for thin objects, boxes, and irregularly shaped items. It also distinguishes the instructed target in cluttered or referential settings. These results demonstrate that RynnBrain 1.1 can jointly reason about object identity, geometry, state, and intended interaction when generating contact-oriented spatial predictions.

5.6 Real-Robot Evaluation

We evaluate RynnBrain-VLA on real robots to answer three questions: (i) can RynnBrain-VLA achieve stable performance on real-world manipulation tasks? (ii) does a stronger embodied foundation model lead to a better VLA policy? and (iii) can joint training on data from heterogeneous robot embodiments provide mutual benefits across embodiments?

5.6.1 Experimental Setup

Evaluation Protocol. We deploy RynnBrain-VLA on three heterogeneous embodiments, as summarized in Table 6 and Figure 8. For each task, we conduct 20 trials with randomized initial object placements and report the final task success rate under a fixed success criterion. Since some long-horizon tasks can be

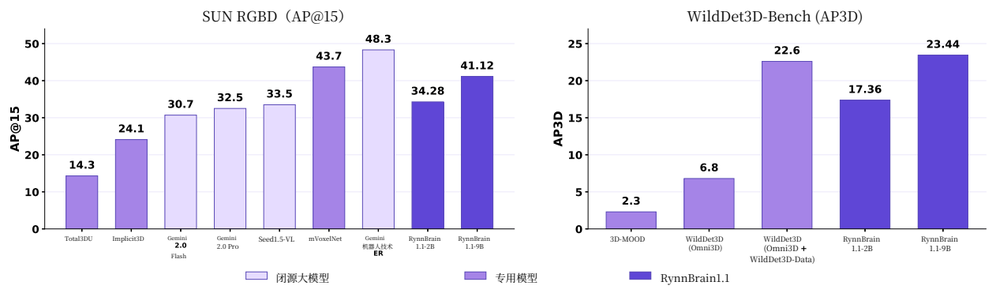

图5 RynnBrain 1.1-2B 与 9B 在 SUN RGB-D 和 WildDet3D-Bench 上的性能比较。

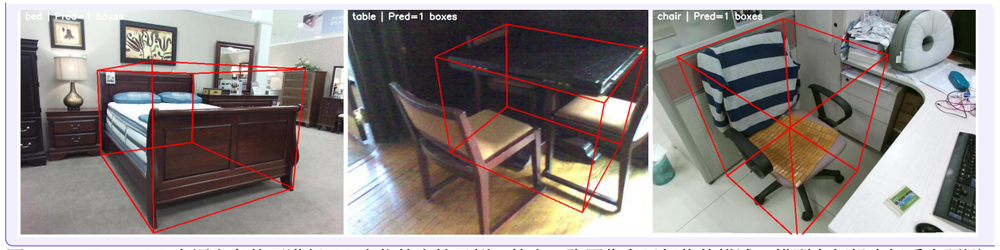

图6 RynnBrain 1.1 在语言条件下进行 3D 定位的定性示例。给定一张图像和目标物体描述，模型在相机坐标系中预测一个有向 3D 边界框。这些示例涵盖了室内环境中具有不同尺度、形状、视角和遮挡程度的物体。

5.5 接触点预测

评估接触点预测仍然具有挑战性，因为目前还没有能够可靠反映预测接触功能有效性的标准化指标。与单一标注的像素级距离可能会惩罚对操作同样有效的其他接触点，而传统抓取框指标无法直接适用于我们以接触为中心的表征。因此，我们在图7中提供定性评估，以展示 RynnBrain 1.1 在多样物体和指令下的能力。这里我们展示由 RynnBrain 1.1-9B 生成的案例。

可视化案例表明，RynnBrain 1.1 能够识别与任务相关的接触位置，而不仅仅是预测物体中心。例如，当被要求拿起杯子时，模型会选择杯柄；当杯子被倒置时，它会调整预测；对于细长物体、盒子和形状不规则的物品，它也能生成合适的接触位置和方向。它还能在杂乱或指代性场景中区分所指示的目标。这些结果表明，RynnBrain 1.1 在生成面向接触的空间预测时，能够联合推理物体身份、几何形状、状态和预期交互。

5.6 真实机器人评估

我们在真实机器人上评估 RynnBrain-VLA，以回答三个问题：(i) RynnBrain-VLA 能否在真实世界操作任务中实现稳定性能？(ii) 更强的具身基础模型是否会带来更好的 VLA 策略？(iii) 能否通过对来自异构机器人本体的数据进行联合训练，在不同本体之间产生互益？

5.6.1 实验设置

评估协议。我们将 RynnBrain-VLA 部署在三个异构具身本体上，如表 6 和图 8 所示。对于每个任务，我们进行 20 次试验，并随机化物体的初始摆放位置，在固定成功判据下报告最终任务成功率。由于某些长时程任务可以被

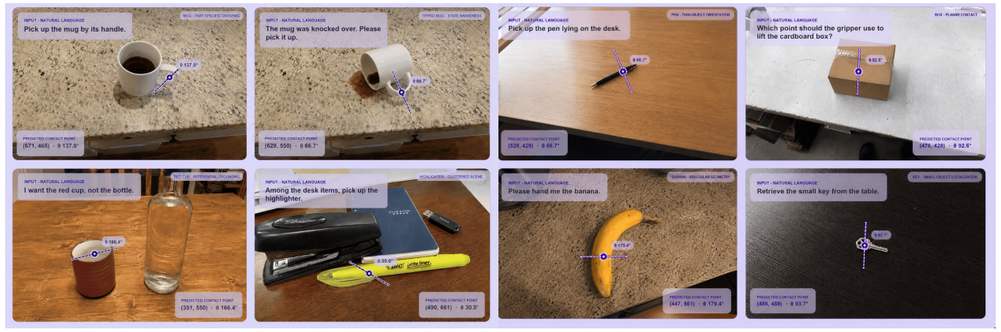

Figure 7 Qualitative results of contact point prediction by RynnBrain 1.1. Given an image and a natural-language instruction, the model predicts an action-relevant contact point together with its in-plane grasp orientation. The examples cover diverse object geometries, object states, target parts, and cluttered scenes, demonstrating instruction-conditioned contact grounding beyond object-level localization. Coordinates are normalized to $[0, 1000]$.

naturally decomposed into multiple sub-tasks, final success alone does not fully reflect the execution progress of a policy. We therefore additionally report a process score that measures the fraction of completed sub-tasks. For a task with K sub-tasks, the process score is defined as

$$\text{ProcessScore} = \frac{1}{20K} \sum_{i=1}^{20} \sum_{j=1}^{K} s_{i,j}, \tag{2}$$

where $s_{i,j} \in \{0, 1\}$ indicates whether the j-th sub-task is successfully completed in the i-th trial. Sprinkle the Petals, Pour the Wine, and Grab the Spatulas contain 4, 5, 3 subtasks, respectively. This metric provides a more fine-grained measure of task progress, especially for long-horizon manipulation tasks in which a policy may complete several intermediate stages without reaching the final goal.

Deployment and Action Space. For all tasks, inference is performed locally on a workstation equipped with a single NVIDIA GeForce RTX 4090 GPU. The workstation is directly connected to the robot and transmits the predicted actions for real-time execution. For tasks involving the Unitree G1 humanoid, we employ SONIC as the whole-body controller. The model predicts a 64-dimensional SONIC latent motion token alongside a 14-dimensional dexterous-hand prediction corresponding to the G1's three-finger-joint hand, which is represented within the unified action space's Hand group. These two outputs are concatenated into a 78-dimensional action representation and decoded by the SONIC controller into target positions for the actuated joints of the Unitree G1. The 64-dimensional SONIC token is a G1-specific output that lies outside the 81-dimensional unified action space and is used only for the Unitree G1. For tasks involving the Tianji-Wuji system (Tianji Arm and Wuji Hand), the policy activates the Arm-Joint group (14D, two arms) and the Hand group (40D, two dexterous hands), yielding a 54-dimensional active action representation that is sent directly to the Tianji-Wuji controller. For tasks involving the Astribot S1, the policy activates the Arm-Joint (14D), Gripper (2D), Head (3D) and Torso (4D), with the dexterous-hand dimensions masked out.

Baselines and Training Settings. We select GR00T N1.7 [48] and $\pi 0.5$ [6] as representative generalist VLA baselines. We additionally construct Qwen-Based-VLA by applying the same VLA post-training recipe used for RynnBrain-VLA to the Qwen base model. For RynnBrain-VLA, we consider two training settings. In the single-task setting, a separate policy is fine-tuned for each individual task. In the generalist setting, we jointly fine-tune a single policy on multi-task and multi-embodiment data using our unified action space, and denote the resulting model as RynnBrain-VLA$_{\text{Generalist}}$. To obtain the best deployment performance for each policy, we use method-specific action-chunk lengths rather than enforcing a shared chunk size across all models. Specifically, we use an action-chunk length of 40 for GR00T N1.7, 50 for $\pi 0.5$, and 32 for RynnBrain-VLA, including both the single-task and generalist settings, and Qwen-Based-VLA.

Tasks. We design three challenging long-horizon manipulation tasks for quantitative comparison: *Sprinkle*

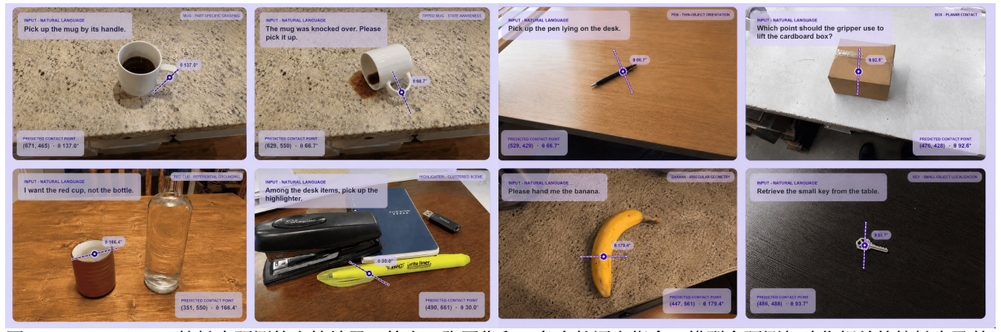

图 7 RynnBrain 1.1 接触点预测的定性结果。给定一张图像和一条自然语言指令，模型会预测与动作相关的接触点及其平面内抓取方向。这些示例涵盖了多样化的物体几何形状、物体状态、目标部件和杂乱场景，展示了超越物体级定位的、以指令为条件的接触点定位。坐标归一化到 $[0, 1000]$。

自然地分解为多个子任务，仅凭最终成功并不能完全反映策略的执行进度。因此，我们还额外报告一个过程得分，用于衡量已完成子任务的比例。对于包含 K 个子任务的任务，过程得分定义为

$$\text{ProcessScore} = \frac{1}{20K} \sum_{i=1}^{20} \sum_{j=1}^{K} s_{i,j}, \tag{2}$$

其中 $s_{i,j} \in \{0, 1\}$ 表示第 j 个子任务是否在第 i 次试验中成功完成。“撒花瓣”、“倒酒”和“抓取锅铲”分别包含 4、5、3 个子任务。该指标提供了更细粒度的任务进度度量，尤其适用于长时程操作任务；在此类任务中，策略可能完成若干中间阶段而尚未达到最终目标。

部署与动作空间。对于所有任务，推理均在配备单块 NVIDIA GeForce RTX 4090 GPU 的工作站上本地执行。该工作站直接连接机器人，并传输预测动作以进行实时执行。对于涉及 Unitree G1 人形机器人的任务，我们采用 SONIC 作为全身控制器。模型会预测一个 64 维 SONIC 潜在运动 token，以及一个 14 维灵巧手预测，对应 G1 的三指关节手；该预测在统一动作空间的 Hand 组中表示。这两个输出拼接成一个 78 维动作表示，并由 SONIC 控制器解码为 Unitree G1 各驱动关节的目标位置。64 维 SONIC token 是 G1 专属输出，不属于 81 维统一动作空间，并且仅用于 Unitree G1。对于涉及 Tianji-Wuji 系统（Tianji 机械臂和 Wuji 手）的任务，策略会激活 Arm-Joint 组（14D，两条机械臂）和 Hand 组（40D，两只灵巧手），生成 54 维的激活动作表示，并直接发送给 Tianji-Wuji 控制器。对于涉及 Astribot S1 的任务，策略会激活 Arm-Joint（14D）、Gripper（2D）、Head（3D）和 Torso（4D），并将灵巧手维度屏蔽。

基线模型与训练设置。我们选择 GR00T N1.7 [48] 和 $\pi 0.5$ [6] 作为具有代表性的通用 VLA 基线。我们还通过将用于 RynnBrain-VLA 的相同 VLA 后训练方案应用于 Qwen 基础模型，构建了 Qwen-Based-VLA。对于 RynnBrain-VLA，我们考虑两种训练设置。在单任务设置中，针对每个单独任务分别微调一个独立策略。在通用设置中，我们使用统一的动作空间，在多任务和多具身数据上联合微调单个策略，并将所得模型记为 RynnBrain-VLA$_{\text{Generalist}}$。为获得每个策略的最佳部署性能，我们采用各方法特定的动作块长度，而不是强制所有模型使用共享的块大小。具体而言，我们对 GR00T N1.7 使用 40 的动作块长度，对 $\pi 0.5$ 使用 50，对 RynnBrain-VLA（包括单任务和通用设置）以及 Qwen-Based-VLA 使用 32。

任务。 我们设计了 th ree 具有挑战性的长时程操作任务，用于定量比较：Sprinkl e

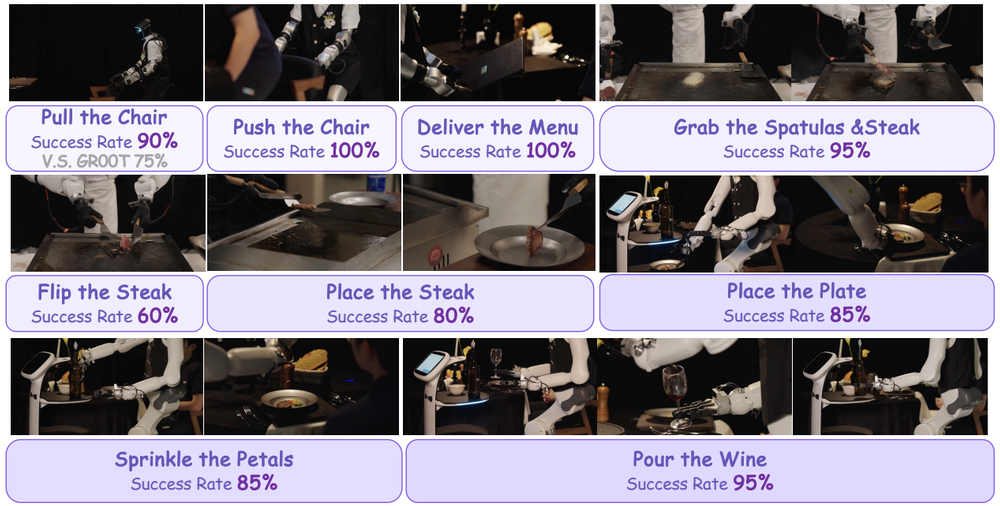

Figure 8 Real-robot evaluation across all tasks considered in our experiments. The tasks are performed in diverse real-life environments and cover whole-body manipulation, bimanual coordination, dexterous-hand operation, object placement, furniture interaction, and object delivery. Representative execution snapshots and success rates over 20 trials are shown for each task.

the Petals, *Pour the Wine*, and *Grab the Spatulas*. Each task consists of multiple temporally ordered subtasks and requires the policy to complete a sequence of perception, reaching, grasping, manipulation, and placement operations. The task setups are illustrated in Figure 8. The first two tasks are performed on the Astribot robot with gripper-based whole-body control, where the robot needs to actively change its viewpoint during execution. In *Sprinkle the Petals*, the policy must continuously observe the container and determine whether the petals have been fully poured out before proceeding, requiring closed-loop visual observation and task-progress understanding. *Pour the Wine* requires the robot to precisely align the bottle opening with the target glass and execute a controlled pouring motion, placing stronger demands on fine-grained affordance understanding and precise manipulation. *Grab the Spatulas* is performed with Tianji-Wuji dexterous hands and requires fine-grained grasping and hand control to reliably manipulate the tools. Together, these tasks cover complementary challenges in long-horizon reasoning, visual feedback, precise affordance-guided manipulation, and dexterous interaction, and serve as the main benchmark for comparing different VLA methods and training settings. Beyond the three main tasks, we further evaluate RynnBrain-VLA on a series of additional robot tasks across heterogeneous embodiments, including object placement, flipping, furniture interaction, and object delivery. These evaluations broaden the coverage of embodiments and manipulation primitives and provide an additional assessment of policy generality.

5.6.2 Stable Real-World Performance

As shown in Figure 8 and Table 6, RynnBrain-VLA achieves stable performance across multiple robot embodiments, including the Astribot bimanual platform, the Tianji-Wuji dexterous-hand system, and the Unitree G1 humanoid. The policy performs reliably under substantially different action spaces and control interfaces, covering gripper-based whole-body manipulation, dexterous grasping, bimanual coordination, and humanoid whole-body control. In particular, the Astribot tasks require the robot to actively change its viewpoint during execution and continuously update its actions from visual feedback. The strong results on these tasks indicate that RynnBrain-VLA remains robust to viewpoint changes rather than relying on a fixed observation configuration.

RynnBrain-VLA also performs well on the Unitree G1 humanoid. On *Pull the Chair*, which involves coordinated humanoid whole-body motion, RynnBrain-VLA achieves a 90% success rate, compared with 75% for

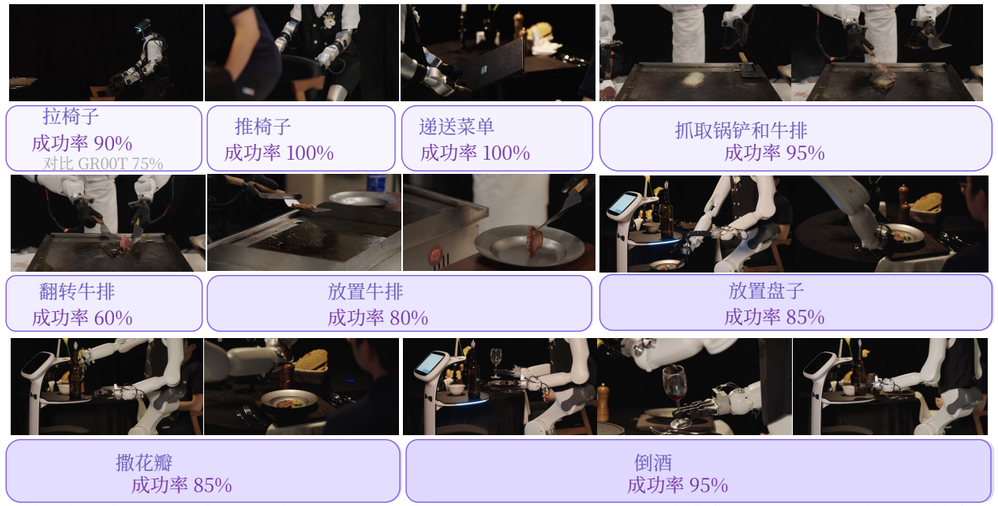

图 8 针对我们实验中考虑的所有任务进行的真机评估。这些任务在多样的真实生活环境中执行，涵盖全身操作、双臂协调、灵巧手操作、物体放置、家具交互和物体递送。每个任务均展示了代表性执行快照以及 20 次试验的成功率。

the Petals、Pour the Wine 和 Grab the Spatulas。每个任务都由多个按时间顺序排列的子任务组成，并要求策略完成一系列感知、伸取、抓取、操作和放置操作。任务设置如图 8 所示。前两个任务在 Astribot 机器人上执行，采用基于夹爪的全身控制，机器人需要在执行过程中主动改变其视角。在 Sprinkle the Petals 中，策略必须持续观察容器，并判断花瓣是否已被完全倒出后才能继续，这要求闭环视觉观察和任务进度理解。Pour the Wine 要求机器人精确地将瓶口与目标酒杯对齐，并执行受控的倾倒动作，这对细粒度可供性理解和精确操作提出了更高要求。Grab the Spatulas 使用 Tianji-Wuji 灵巧手执行，需要细粒度抓取和手部控制，以可靠地操作工具。这些任务共同涵盖了长时程推理、视觉反馈、精确的可供性引导操作以及灵巧交互中的互补挑战，并作为比较不同 VLA 方法和训练设置的主要基准。除这三个主要任务外，我们还在跨异构本体的一系列附加机器人任务上进一步评估 RynnBrain-VLA，包括物体放置、翻转、家具交互和物体递送。这些评估拓宽了本体和操作原语的覆盖范围，并为策略泛化性提供了额外评估。

5.6.2 稳定的真实世界性能

如图 8 和表 6 所示，RynnBrain-VLA 在多种机器人本体上均取得了稳定性能，包括 Astribot 双臂平台、Tianji-Wuji 灵巧手系统以及 Unitree G1 人形机器人。该策略在差异显著的动作空间和控制接口下仍能可靠运行，涵盖基于夹爪的全身操作、灵巧抓取、双臂协调以及人形机器人全身控制。特别地，Astribot 任务要求机器人在执行过程中主动改变视角，并持续根据视觉反馈更新动作。这些任务上的优异结果表明，RynnBrain-VLA 对视角变化保持稳健，而非依赖固定的观测配置。

RynnBrain-VLA 在 Unitree G1 人形机器人上也表现出色。在涉及人形机器人全身协调运动的 Pull the Chair 任务中，RynnBrain-VLA 取得了 90% 的成功率，而 75% 的成功率则来自

Table 6 Real-robot evaluation results. Each task reports the process score and final success rate (%). The process score measures the average fraction of completed sub-tasks, while a trial is counted as successful only when all sub-tasks of the corresponding task are completed.

Model	Sprinkle the Petals *Astribot*		Pour the Wine *Astribot*		Grab the Spatulas *Tianji-Wuji*		Average –	
	Proc.	**Succ.**	**Proc.**	**Succ.**	**Proc.**	**Succ.**	**Proc.**	**Succ.**
Qwen-Based-VLA	75.00	65.00	75.00	65.00	55.00	50.00	68.33	60.00
GR00T N1.7	86.25	75.00	77.00	65.00	86.67	80.00	83.31	73.33
$\pi_{0.5}$	65.00	60.00	69.00	65.00	83.33	70.00	72.44	65.00
RynnBrain-VLA	87.50	**85.00**	88.00	80.00	**98.33**	**95.00**	91.28	86.67
RynnBrain-VLA$_{\text{Generalist}}$	**88.75**	**85.00**	**97.00**	**95.00**	96.67	**95.00**	**94.14**	**91.67**

GR00T N1.7 combined with the same SONIC controller. Since GR00T N1.7 with SONIC is a natural baseline for predicting SONIC motion tokens, this comparison provides additional evidence that RynnBrain-VLA can produce reliable actions even for challenging whole-body control tasks. Overall, the results demonstrate that its real-world performance is stable not only across tasks, but also across robot morphologies, viewpoints, and control configurations.

5.6.3 A Stronger Embodied Base Yields a Better VLA

We next investigate whether an embodied foundation model provides a stronger initialization for VLA post-training. RynnBrain-VLA and Qwen-Based-VLA follow the same VLA post-training recipe and use the same action-chunk length, while being initialized from RynnBrain and Qwen Based Model, respectively. As shown in Table 6, RynnBrain-VLA consistently outperforms Qwen-Based-VLA across all three long-horizon tasks. Its average process score improves from 68.33% to 91.28%, while the average final success rate increases from 60.00% to 86.67%. The advantage is particularly pronounced on *Grab the Spatulas*, where RynnBrain-VLA achieves a 95% success rate compared with 50% for Qwen-Based-VLA. These results suggest that the capabilities acquired during embodied foundation-model training transfer effectively to downstream VLA post-training. Compared with directly post-training a general-purpose VLM as a VLA policy, initializing from RynnBrain leads to more reliable execution of multi-stage manipulation tasks and substantially fewer failures during intermediate stages. This comparison provides direct evidence that a stronger embodied base model can serve as a better foundation for real-world VLA policies.

5.6.4 Cross-Embodiment Generalization

To examine whether the capability of RynnBrain-VLA is tied to a specific robot embodiment, we adapt it to three morphologically and mechanically distinct robot platforms through the deployment framework described in Section 4.3. As shown in Table 6 and Figure 8, RynnBrain-VLA achieves stable performance across humanoid robots, dexterous-hand manipulation, whole-body control, and bimanual grasping settings. These embodiments differ substantially in morphology, action dimensionality, and low-level control interfaces. The consistent performance across these settings suggests that the embodied representations learned by RynnBrain capture task- and interaction-level knowledge that is not tightly coupled to a particular robot morphology or action parameterization, allowing the same embodied foundation model to serve as a shared starting point for heterogeneous robot policies.

More interestingly, we find that jointly training multiple tasks and embodiments within a single VLA does not introduce obvious interference. With the unified action-space strategy in Section 4.2, RynnBrain-VLA$_{\text{Generalist}}$ improves the average process score from 91.28% to 94.14% and the final success rate from 86.67% to 91.67% compared with separately fine-tuned single-task policies. We hypothesize that different embodiments share common interaction structures—such as object affordances, task progress, and reach–grasp–manipulate patterns—even though their executable actions are different. Joint training therefore allows these shared structures to reinforce a common visual–action representation, while unified action-space

表 6 真实机器人评估结果。每项任务报告过程得分和最终成功率（%）。过程得分衡量已完成子任务的平均比例，而仅当对应任务的所有子任务均完成时，一次试验才计为成功。

模型	撒花瓣 *Astribot*		倒酒 *Astribot*		抓取锅铲 *Tianji-Wuji*		平均值 –	
	过程	**成功率**	**过程**	**成功**	**过程**	**成功**	**过程**	**成功**
基于 Qwen 的 VLA	75.00	65.00	75.00	65.00	55.00	50.00	68.33	60.00
GR00T N1.7	86.25	75.00	77.00	65.00	86.67	80.00	83.31	73.33
$\pi_{0.5}$	65.00	60.00	69.00	65.00	83.33	70.00	72.44	65.00
RynnBrain-VLA	87.50	**85.00**	88.00	80.00	**98.33**	**95.00**	91.28	86.67
RynnBrain-VLA$_{\text{Generalist}}$	**88.75**	**85.00**	**97.00**	**95.00**	96.67	**95.00**	**94.14**	**91.67**

GR00T N1.7 与同一 SONIC 控制器结合。由于 GR00T N1.7 结合 SONIC 是预测 SONIC 运动 token 的自然基线，这一比较提供了额外证据，表明 RynnBrain-VLA 即使在具有挑战性的全身控制任务中也能生成可靠动作。总体而言，结果表明，其真实世界性能不仅在任务之间保持稳定，在机器人形态、视角和控制配置之间也同样稳定。

5.6.3 更强的具身基座模型带来更好的 VLA

我们接下来研究，具身基础模型能否为 VLA 后训练提供更强的初始化。RynnBrain-VLA 与 Qwen-Based-VLA 遵循相同的 VLA 后训练方案，并使用相同的动作块长度，但分别从 RynnBrain 和 Qwen Based Model 初始化。如表 6 所示，RynnBrain-VLA 在全部三个长时程任务上均持续优于 Qwen-Based-VLA。其平均过程得分从 68.33% 提升至 91.28%，平均最终成功率从 60.00% 提升至 86.67%。这一优势在 Grab the Spatulas 上尤为明显：RynnBrain-VLA 达到 95% 的成功率，而 Qwen-Based-VLA 为 50%。这些结果表明，在具身基础模型训练期间获得的能力能够有效迁移到下游 VLA 后训练中。与直接将通用 VLM 后训练为 VLA 策略相比，从 RynnBrain 初始化可使多阶段操作任务执行更可靠，并显著减少中间阶段的失败。这一对比提供了直接证据，表明更强的具身基座模型可以作为真实世界 VLA 策略更好的基础。

5.6.4 跨具身泛化

为了检验 RynnBrain-VLA 的能力是否与特定的机器人具身绑定，我们通过第 4.3 节所述的部署框架将其适配到三个在形态和机械结构上显著不同的机器人平台。如表 6 和图 8 所示，RynnBrain-VLA 在人形机器人、灵巧手操作、全身控制以及双臂抓取场景中均取得了稳定性能。这些具身在形态、动作维度和底层控制接口方面存在显著差异。上述场景下的一致性能表明，RynnBrain 学习到的具身表征捕捉到了任务层面和交互层面的知识，而这些知识并未与特定的机器人形态或动作参数化紧密耦合，从而使同一个具身基础模型能够作为异构机器人策略的共享起点。

更有趣的是，我们发现，在单个 VLA 中联合训练多个任务和多种具身本体并不会引入明显的干扰。采用第 4.2 节中的统一动作空间策略，与分别微调的单任务策略相比，RynnBrain- VLA$_{\text{Generalist}}$ 将平均过程得分从 91.28% 提升至 94.14%，并将最终成功率从 86.67% 提升至 91.67%。我们假设，不同的具身本体共享共同的交互结构——例如物体可供性、任务进展以及伸手—抓取—操作模式——尽管它们可执行的动作不同。因此，联合训练使这些共享结构能够强化共同的视觉—动作表征，同时统一动作空间

preserves embodiment-specific control. This suggests that cross-embodiment data can act as complementary supervision: the key benefit of a unified VLA may lie not only in supporting more robots with one model, but also in allowing different embodiments to teach each other how the physical world can be interacted with.

6 Conclusion and Future Work

We presented **RynnBrain 1.1**, an embodied foundation model family spanning 2B, 9B, and 122B-A10B scales. RynnBrain 1.1 studies embodied capability scaling under a unified training framework and demonstrates clear improvements in multi-view reasoning, spatial grounding, and robot-oriented understanding. We further introduce explicit 3D-grounded supervision for the 2B and 9B models, enabling native language-conditioned 3D grounding, together with a compact contact-centered representation for interaction grounding.

We also develop RynnBrain-VLA to connect embodied understanding with real-world action. Under the same post-training recipe, RynnBrain-VLA substantially outperforms a Qwen-based VLA on long-horizon manipulation tasks. Through a unified action space with embodiment-specific masking, the policy is deployed across a Unitree G1 humanoid, an Astribot bimanual robot, and a Tianji-Wuji dexterous-hand system. Joint multi-task and multi-embodiment training further improves real-robot performance, providing initial evidence that heterogeneous robot data can contribute complementary supervision to a shared policy.

Looking forward, we aim to extend RynnBrain toward a unified multimodal model that combines perception, understanding, generation, spatial grounding, and action within a common framework. An important direction is to develop RynnBrain into the cognitive core of a general embodied agent, integrating long-term memory, world modeling, task planning, active perception, and closed-loop interaction. Such an agent could continuously reason about changes in the physical world, generate and revise plans from feedback, and acquire new skills through interaction across diverse tasks and robot embodiments.

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保持各具身本体的特有控制。这表明，跨具身数据可以作为补充监督：统一 VLA 的关键优势可能不仅在于用一个模型支持更多机器人，还在于让不同具身本体相互教会彼此如何与物理世界交互。

6 结论与未来工作

我们提出了 RynnBrain 1.1，这是一个涵盖 2B、9B 和 122B-A10B 规模的具身基础模型系列。RynnBrain 1.1 在统一的训练框架下研究具身能力的规模化，并在多视角推理、空间接地和面向机器人的理解方面展现出明显提升。我们进一步为 2B 和 9B 模型引入显式 3D 接地监督，使其具备原生语言条件三维接地能力，同时引入一种用于交互接地的紧凑接触中心表征。

我们还开发了 RynnBrain-VLA，将具身理解与真实世界动作相连接。在相同的后训练配方下，RynnBrain-VLA 在长时程操作任务上显著优于基于 Qwen 的 VLA。通过带有本体特定掩码的统一动作空间，该策略被部署到 Unitree G1 人形机器人、Astribot 双臂机器人和 Tianji-Wuji 灵巧手系统上。联合多任务与多本体训练进一步提升了真实机器人性能，初步证明异构机器人数据可为共享策略提供互补监督。

展望未来，我们希望将 RynnBrain 扩展为一个统一的多模态模型，在共同框架内融合感知、理解、生成、空间定位与行动。一个重要方向是将 RynnBrain 发展为通用具身智能体的认知核心，整合长期记忆、世界建模、任务规划、主动感知和闭环交互。这样的智能体能够持续推理物理世界中的变化，根据反馈生成并修正计划，并通过跨多种任务和机器人具身形态的交互获取新技能。

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